MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Outotec
LARSEN & TOUBRO

Project: Mansourah Massarah Gold Project

Customer: MA'ADEN GOLD AND BASE METALS (MA'ADEN)

Title: CALCULATION

3120 FOUNDATION DESIGN OF CONVEYOR 3120-FDR-002

DESIGN CALCULATION

AS BUILT

MD-216-3120-EG-CV-CAL-1006

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Global Project ID	**900707**
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Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

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REFERENCE DOCUMENTS

DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria
SBC 301 :2007	Saudi Structural - Loading & Forces Code
SBC 304 :2007	Saudi Code for Concrete Structures

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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities.
RL	:	Reduced Level
C.L.	:	Centre Line
CJ	:	Construction Joint
FDN	:	Foundation
TOC	:	Top of concrete
FGL	:	Finished Ground Level
FFL	:	Finished Floor Level
BOF	:	Bottom of footing
TYP	:	Typical
TOC	:	Top of concrete
IP	:	Insert Plate
RCC	:	Reinforced Cement Concrete
MKD.	:	Marked
THK.	:	Thick

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1. INTENT

The document is prepared for foundation design calculation of Conveyor Gallery 3120-FDR-002.
This conveyor is between TP-2 and Screen house and belt width of the conveyor is 1400 mm.

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.
MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.
The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

This document is prepared for foundation design of Conveyor 3120-FDR-002. All the loads applied to the structure for analysis & design followed Civil -Structural Design Criteria and SBC 301:2007
The column and beam reinforcement design have been performed in Staad-Pro Software. The footings of Trestle TR1 & TR2 have been performed in MS Excel.
Footing of TR1 & TR2 have been analized by FEM analysis in Staad-Pro Software.

4. CONCRETE STRUCTURE

4.1 Design

Design of concrete structures and foundations shall be in accordance with SBC-304 and ACI 318M.

4.2 Materials

Concrete Strength of 28 days
· Foundation and Structural Members: 35 MPa
· Grade slab and pavement: 30 MPa
· Blinding Concrete below foundation bases: 15 Mpa

Reinforcement Grade
· Deformed Bar : ASTM A 615 / A 615M Grade 60 (fy = 420MPa)
· Welded Wire Fabric : ASTM A 185(Plain) / A 497(Deformed)

4.3 Foundations for Structures

Foundation shall be Isolated and combined Shallow Concrete foundation. The design calculations are prepared as per recommendation of standard SBC-304 Clause 13.3.

4.4 Minimum Concrete Cover

Items	Minimum Cover(mm)
Foundation (Side & Bottom)	75
Foundation (Top)	50
Column & Pedestal	50
Beam	50
Slab	40

4.5 Allowable Soil Bearing Capacity

The net safe bearing capacity of soil for foundations design has been considered as 500 kN/m2 at 2m depth below FGL (ie. RL 910.500m) refer Table 4 - BH-17,18,19,25,29 of Geotechnical Report (MD-216-3000-EG-CV-RPT-1008).

4.6 Reference

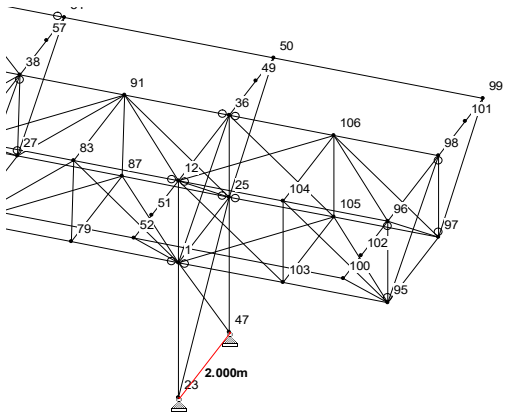
Reference is made to codes, standards and specifications as indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.1.

5 INPUT LOADS FROM SUPERSTRUCTURE

For load calculation of superstructure refer Document No: MD-216-3120-EG-ST-CAL-1006

5.1 Support Reaction from TR1

Node	L/C	Horizontal Fx kN	Vertical Fy kN	Horizontal Fz kN
23	1 DEAD LOAD	0.0	50.4	-8.0
	2 LIVE LOAD	0.0	33.9	-4.0
	3 MATERIAL LOAD	0.0	25.6	-2.8
	4 WL (+Z)	0.0	31.6	-9.9
	5 WL (-Z)	0.0	-31.6	9.9
	6 WL (+X)	0.0	0.0	0.0
	7 WL (-X)	0.0	0.0	0.0
	8 TL1	0.0	-0.1	22.4
	9 TL2	0.0	0.1	-22.4
	10 EL (+X)	-2.0	0.8	-0.1
	11 EL (+Z)	0.0	4.1	-1.2
47	1 DEAD LOAD	0.0	43.2	2.7
	2 LIVE LOAD	0.0	34.1	4.0
	3 MATERIAL LOAD	0.0	25.8	2.8
	4 WL (+Z)	0.0	-31.6	-10.5
	5 WL (-Z)	0.0	31.6	10.5
	6 WL (+X)	0.0	0.0	0.0
	7 WL (-X)	0.0	0.0	0.0
	8 TL1	0.0	0.1	-22.4
	9 TL2	0.0	-0.1	22.4
	10 EL (+X)	-2.0	0.8	0.1
	11 EL (+Z)	0.0	-4.1	-1.2



Staad model of TR1 superstructure

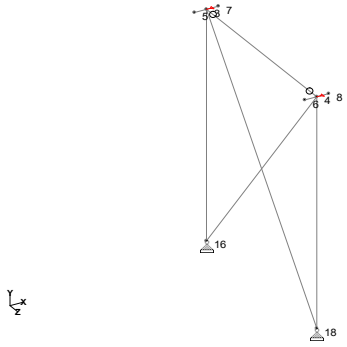
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5.2 Support Reaction from TR2

Node	L/C	Horizontal Fx kN	Vertical Fy kN	Horizontal Fz kN
16	1 DEAD LOAD	0.6	51.4	0.7
	2 LIVE LOAD	0.2	32.1	2.4
	3 MATERIAL LOAD	0.1	25.6	1.4
	4 WL (+Z)	0.0	-43.6	-12.7
	5 WL (-Z)	0.0	43.7	12.7
	6 WL (+X)	-0.3	0.0	0.0
	7 WL (-X)	0.3	0.0	0.0
	8 TL1	0.0	-0.5	7.4
	9 TL2	0.0	0.5	-7.4
	10 EL (+X)	-0.1	0.5	0.0
	11 EL (+Z)	0.0	-8.5	-2.7
18	1 DEAD LOAD	0.0	49.8	-7.1
	2 LIVE LOAD	0.3	31.9	-2.4
	3 MATERIAL LOAD	0.2	25.4	-1.4
	4 WL (+Z)	0.0	43.6	-13.0
	5 WL (-Z)	0.0	-43.7	13.0
	6 WL (+X)	-0.3	0.0	0.0
	7 WL (-X)	0.3	0.0	0.0
	8 TL1	0.0	0.5	-7.4
	9 TL2	0.0	-0.5	7.4
	10 EL (+X)	-0.1	0.5	0.0
	11 EL (+Z)	0.0	8.5	-2.8



Staad model of TR2 superstructure

5.3 LOAD COMBINATIONS

5.3.1 Load Combination for Base pressure calculation

(As per Section 2.4 of SBC 301 -2007)

Load Cases	Reference
LOAD COMB 301 1.0 DL	(Eq. 2.4.1-1 of SBC 301)
LOAD COMB 302 1.0 (DL + TL1) + 1.0 (LL+ML)	(Eq. 2.4.1-2 of SBC 301)
LOAD COMB 303 1.0 (DL + TL1) + 1.0 (LL+ML)	
LOAD COMB 304 1.0 DL+ 0.75 WL(+Z) + 0.75 (LL+ML)	(Eq. 2.4.1-6 of SBC 301)
LOAD COMB 305 1.0 DL+ 0.75 WL(-Z) + 0.75 (LL+ML)	
LOAD COMB 306 1.0 DL+ 0.75 WL(+X) + 0.75 (LL+ML)	
LOAD COMB 307 1.0 DL+ 0.75 WL(-X) + 0.75 (LL+ML)	
LOAD COMB 308 1.0 DL + 0.525 EL(+X) + 0.75 (LL+ML)	
LOAD COMB 309 1.0 DL + 0.525 EL(-X) + 0.75 (LL+ML)	
LOAD COMB 310 1.0 DL + 0.525 EL(+Z) + 0.75 (LL+ML)	
LOAD COMB 311 1.0 DL + 0.525 EL(-Z) + 0.75 (LL+ML)	
LOAD COMB 312 0.6 DL + 1.0 WL(+Z)	(Eq. 2.4.1-7 of SBC 301)
LOAD COMB 313 0.6 DL + 1.0 WL(-Z)	
LOAD COMB 314 0.6 DL + 1.0 WL(+X)	
LOAD COMB 315 0.6 DL + 1.0 WL(-X)	
LOAD COMB 316 0.6 DL + 0.7 EL(+X)	(Eq. 2.4.1-8 of SBC 301)
LOAD COMB 317 0.6 DL + 0.7 EL(-X)	
LOAD COMB 318 0.6 DL + 0.7 EL(+Z)	
LOAD COMB 319 0.6 DL + 0.7 EL(-Z)	

5.3.2 Load Combination for Concrete design

Load Case	Reference
LOAD COMB 401 1.4 DL	(Eq. 2.3.2-1 of SBC 301)
LOAD COMB 402 1.4 (DL + TL1) + 1.7 (LL+ML)	(Eq. 2.3.2-2 of SBC 301)
LOAD COMB 403 1.4 (DL + TL1) + 1.7 (LL+ML)	
LOAD COMB 404 1.2 DL+ 1.6 WL(+Z) + 0.5 (LL+ML)	(Eq. 2.3.2-4 of SBC 301)
LOAD COMB 405 1.2 DL+ 1.6 WL(-Z) + 0.5 (LL+ML)	
LOAD COMB 406 1.2 DL+ 1.6 WL(+X) + 0.5 (LL+ML)	
LOAD COMB 407 1.2 DL+ 1.6 WL(-X) + 0.5 (LL+ML)	
LOAD COMB 408 1.2 DL + 1.0 EL(+X) + 0.5 (LL+ML)	(Eq. 2.3.2-5 of SBC 301)
LOAD COMB 409 1.2 DL + 1.0 EL(-X) + 0.5 (LL+ML)	
LOAD COMB 410 1.2 DL + 1.0 EL(+Z) + 0.5 (LL+ML)	
LOAD COMB 411 1.2 DL + 1.0 EL(-Z) + 0.5 (LL+ML)	
LOAD COMB 412 0.9 DL + 1.6 WL(+Z)	(Eq. 2.3.2-6 of SBC 301)
LOAD COMB 413 0.9 DL + 1.6 WL(-Z)	
LOAD COMB 414 0.9 DL + 1.6 WL(+X)	
LOAD COMB 415 0.9 DL + 1.6 WL(-X)	
LOAD COMB 416 0.9 DL + 1.0 EL(+X)	(Eq. 2.3.2-7 of SBC 301)
LOAD COMB 417 0.9 DL + 1.0 EL(-X)	
LOAD COMB 418 0.9 DL + 1.0 EL(+Z)	
LOAD COMB 419 0.9 DL + 1.0 EL(-Z)	

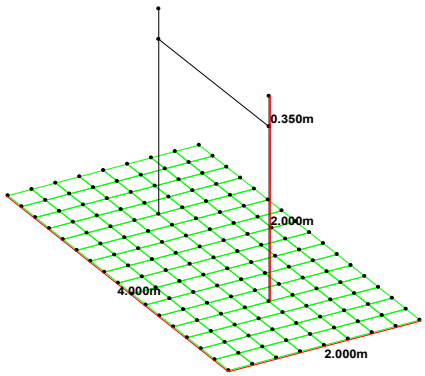
Eq. 2.3.2-3 of SBC 301 - 2007 has not been considered as it would not produce worst load combination in the absence of Roof Live Load & Rain Load

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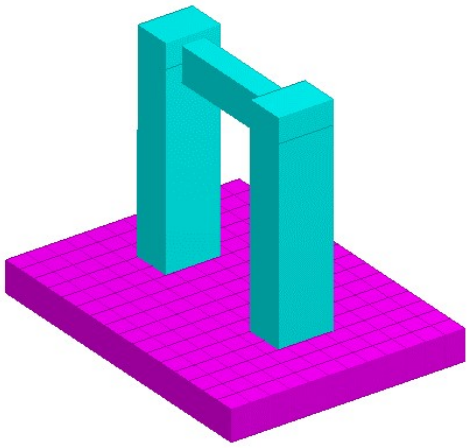
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6 GEOMETRY

6.1 Geometry of Combined Footing mkd. CF1 for Trestle TR1 & TR2



Staad Model of TR1 & TR2 Foundation mkd CF1 (Along with dimension)



Isometric view of Foundation mkd CF1

7. DESIGN OF CONCRETE COMPONENTS

All footings have been designed using worksheet in accordance to SBC-304.
All beams & columns have been designed in Staad-Pro by ACI 318M.
For Design Results, Refer Attachment 3 & 4 - Staad Output file.

7.1 Design Summary of Pedestal

Column mark	Width (Along conv)	Depth (Across conv)	Required Steel (mm^2)	Reinforcement	Provided Steel (mm^2)	Links	Remarks	Numbers
P1	600	600	3600	12 No.- D20	3768	3 Nos D10 @ 200c/c	For TR1 & TR2	4

7.2 Design Summary of Beam

Beam mark	Width (mm)	Depth (mm)	Top Reinforcement	Bottom Reinforcement	Stirrup	Numbers
TB1	300	500	3 Nos D16	3 Nos D16	2L D10 STRPS @ 200c/c	2

7.3 Design Summary of Foundation

Structure Name	Foundation mark	Length (Along conv) (mm)	Width (Across conv) (mm)	Thickness (mm)	Max pressure (kN/m^2)	Top reinforcement	Bottom reinforcement	F.O.S.	Numbers
TR1	CF1	2000	4000	400	111	D12 @ 200c/c B/W	D16 @ 200c/c B/W	0.20	1
TR2	CF1	2000	4000	400	89	D12 @ 200c/c B/W	D16 @ 200c/c B/W	0.16	1

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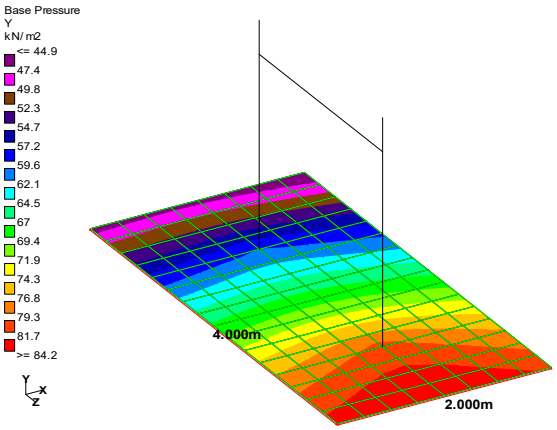
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7.3.1 Design of combined footing mkd CF1 for Trestle mkd. TR1

For Calculation of Base pressure and design of beam, column and raft, Refer STAAD File: **3120-FDR-002_TR1 FDN**

Staad Supoort Nodes are 5 & 6



Finished Floor level	F.F.L	=	0.0	m	(Corresponds to RL +913.0m)
Finished Ground level	F.G.L	=	-0.5	m	
Bottom of foundation	B.O.F	=	-2.5	m	
Top of pedestal (staad support)	T.O.F	=	-0.15	m	
Ground Water Level	G.W.L	=	-10	m	
Length of footing	Lz	=	2.000	m	
Width of footing	Lx	=	4.000	m	
Thickness of footing	D	=	0.40	m	
Depth of pedestal	Bx	=	0.60	m	
Width of pedestal	Bz	=	0.60	m	
Unit wt of soil		=	18.0	KN/m^3	
Allowable Net Bearing Capacity		=	500	KN/m^2	
Gross Bearing Capacity		=	536	KN/m^2	
Soil load on footing in staad		=	28.8	KN/m^2	staad input

1) Check for Gross Base Pressure :

Maximum Base Pressure		=	84.2	KN/m^2	Load Case 304
Minimum Base Pressur		=	10.4	KN/m^2	Load Case 312

2) Design of foundation Raft :

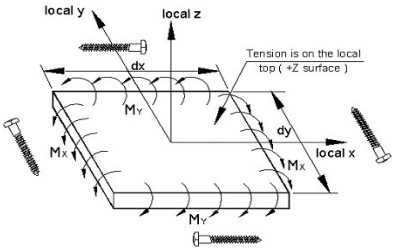
Slab Thickness =	400 mm	Yield Strength of Main Reinforcement , fy =	420 N/mm^2
Width =	1000 mm	Yield Strength of Shear Reinforcement , fyt =	420 N/mm^2
Clear Cover =	75 mm	Strength Reduction Factor of Main Reinforcement , Φ =	0.9
Compressive Strength of Concrete , fc' =	35 N/mm^2	trength Reduction Factor of Shear Reinforcement , Φt =	0.75

STAAD OUTPUT FROM FACTORED LOAD COMBINATION :

(Staad Group : MX,MY)

MOMENT	VALUE	UNIT	TORSIONAL MOMENT	VALUE	UNIT	
Max. Mx	1.601	kN-m	Corresponding Mxy	-1.243	kN-m	*(Plate No. - 112 , L/C No. - 412)*
Min. Mx	-11.576	kN-m	Corresponding Mxy	-4.769	kN-m	*(Plate No. - 27 , L/C No. - 402)*
Max. My	7.003	kN-m	Corresponding Mxy	-0.589	kN-m	*(Plate No. - 113 , L/C No. - 412)*
Min. My	-17.072	kN-m	Corresponding Mxy	-1.759	kN-m	*(Plate No. - 26 , L/C No. - 404)*

FOR TOP FACE REINFORCEMRNT :



Mx1 =	Mx + abs(Mxy) =	1.601 + 1.243 =	2.844 kN-m
My1 =	My + abs(Mxy) =	7.003 + 0.589 =	7.592 kN-m
Mx2 =	Mx+abs(Mxy^2/My) =	1.601 + 0.221 =	1.822 kN-m
My2 =	My+abs(Mxy^2/Mx) =	7.003 + 0.123 =	7.126 kN-m

DESIGN MOMENT CALCULATION FOR TOP FACE REINFORCEMENT :

Mxd = 2.844 kN-m
Myd = 7.592 kN-m

ALONG LOCAL X - DIRECTION :

Mu = 2.844 kN-m
Φ*Kn = Mu/bd^2
Mu/bd^2 = 2.844X10^6/(1000X319^2) = 0.028 N/mm^2 < 6.9 N/mm^2
deff. = 319 mm
Pt = 0.0001 Reqd. Ast = 0.0001X1000X319/100 = 0.32 cm^2
Provided 12 mm @ 200 C/C (Provide Ast = 5.65)cm^2

Total Provide Ast = 5.65 cm^2 O.K.
Total Provide Pt = 0.0018

ALONG LOCAL Y - DIRECTION :

Mu = 7.592 kN-m
Φ*Kn = Mu/bd^2
Mu/bd^2 = 7.592X10^6/(1000X307^2) = 0.075 N/mm^2 < 6.9 N/mm^2
deff. = 307 mm
Pt = 0.0002 Reqd. Ast = 0.0002X1000X319/100 = 0.61 cm^2
Provided 12 mm @ 200 C/C (Provide Ast = 5.65)cm^2

Total Provide Ast = 5.65 cm^2 O.K.
Total Provide Pt = 0.0018

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
Date	16-04-2020	13-05-2020	31.10.2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	APR	APR	APR
Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2
Date	16-04-2020	13-05-2020	31.10.2023
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Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI

FOR BOTTOM FACE REINFORCEMRNT :

Mx1 = Mx - abs(Mxy) = -11.576 - 4.769 = -16.345 kN-m
My1 = My - abs(Mxy) = -17.072 - 1.759 = -18.831 kN-m
Mx2 = Mx-abs(Mxy^2/My) = -11.576 - 1.332 = -12.908 kN-m
My2 = My-abs(Mxy^2/Mx) = -17.072 - 0.267 = -16.805 kN-m

DESIGN MOMENT CALCULATION FOR BOTTOM FACE REINFORCEMENT :

Mxd = -16.345 kN-m
Myd = -18.831 kN-m

ALONG LOCAL X - DIRECTION :

Mu = 16.345 kN-m
Φ*Kn = Mu/bd^2
Mu/bd^2 = 16.345X10^6/(1000X317^2) = 0.163 N/mm^2 < 6.9 N/mm^2
deff. = 317 mm
Pt = 0.0004 Reqd. Ast = 0.0004X1000X317/100 = 1.27 cm^2
Provided 16 mm @ 200 C/C (Provide Ast = 10.05)cm^2

Total Provide Ast = 10.05 cm^2 O.K.
Total Provide Pt = 0.0032

ALONG LOCAL Y - DIRECTION :

Mu = 18.831 kN-m
Φ*Kn = Mu/bd^2
Mu/bd^2 = 18.831X10^6/(1000X301^2) = 0.208 N/mm^2 < 6.9 N/mm^2
deff. = 301 mm
Pt = 0.0006 Reqd. Ast = 0.0006X1000X301/100 = 1.81 cm^2
Provided 16 mm @ 200 C/C (Provide Ast = 10.05)cm^2

Total Provide Ast = 10.05 cm^2 O.K.
Total Provide Pt = 0.0033

SHEAR CHECK

(Staad Group : SHEAR)
(FROM FACTORED LOAD COMBINATION)

ALONG LOCAL X - DIRECTION :

SQx = -0.026 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 166 , L/C No. - 404)*
Vu = 8.24 kN

Provided Pt = 0.0032
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0032X8.242X10^3X317/16.345/10^6)X1000X317/10^3
= 302.82 kN
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0032)X1000X317/10^3
= 317.308 kN
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X5.92X1000X317/10^3
= 543.865 kN
HENCE, Vc = 302.82 kN
Φt*Vc = 0.75X302.82 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 227.115 kN **> 8.242T , Hence O.K.**

ALONG LOCAL Y - DIRECTION :

SQy = -0.033 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 74 , L/C No. - 403)*
Vu = 9.933 kN

Pt provided = 0.0033
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0033X9.933X10^3X301/18.831/10^3)X1000X301/10^3
= 287.599 kN
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0033)X1000X301/10^3
= 301.805 KN
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X5.92X1000X301/10^3
= 516.415 kN
HENCE, Vc = 287.599 kN
Φt*Vc = 0.75X287.599 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 215.699 kN **> 9.933T , Hence O.K.**

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
Date	16-04-2020	13-05-2020	31.10.2023
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Approved	PRI	PRI	PRI

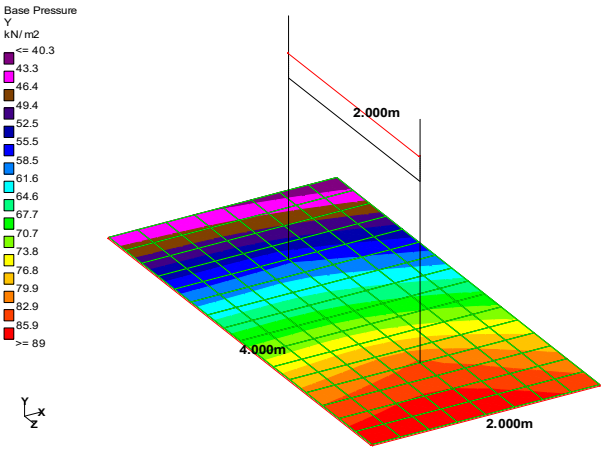
Outotec / L&T Revision No.:	0	1	2
Date	16-04-2020	13-05-2020	31.10.2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
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Approved	PRI	PRI	PRI

7.3.2 Design of combined footing mkd CF1 for Trestle mkd. TR2

For Calculation of Base pressure and design of beam, column and raft, Refer STAAD File: **3120-FDR-002_TR2 FDN**

Staad Supoort Nodes are 5 & 6

Finished Floor level	F.F.L	=	0.0	m	(Corresponds to RL +913.0m)
Finished Ground level	F.G.L	=	-0.5	m	
Bottom of foundation	B.O.F	=	-2.5	m	
Top of pedestal (staad support)	T.O.F	=	-0.15	m	
Ground Water Level	G.W.L	=	-10	m	
Length of footing	Lz	=	2.000	m	
Width of footing	Lx	=	4.000	m	
Thickness of footing	D	=	0.40	m	
Depth of pedestal	Bx	=	0.60	m	
Width of pedestal	Bz	=	0.60	m	
Unit wt of soil		=	18.0	KN/m^3	
Allowable Net Bearing Capacity		=	500	KN/m^2	
Gross Bearing Capacity		=	536	KN/m^2	
Soil load on footing in staad		=	28.8	KN/m^2	staad input



1) Check for Gross Base Pressure :

Maximum Base Pressure	=	89.0	KN/m^2	Load Case	304
Minimum Base Pressur	=	4.5	KN/m^2	Load Case	312

2) Design of foundation Raft :

Slab Thickness =	400	mm	Yield Strength of Main Reinforcement , fy =	420	N/mm^2
Width =	1000	mm	Yield Strength of Shear Reinforcement , fyt =	420	N/mm^2
Clear Cover =	75	mm	Strength Reduction Factor of Main Reinforcement , Φ =	0.9	
Compressive Strength of Concrete , fc' =	35	N/mm^2	Strength Reduction Factor of Shear Reinforcement , Φt =	0.75	

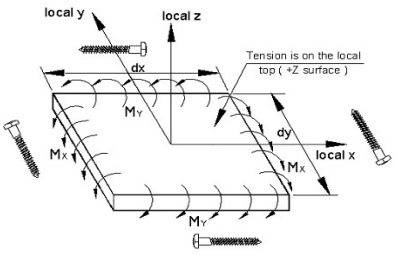
STAAD OUTPUT FROM FACTORED LOAD COMBINATION :

(Staad Group : MX,MY)

MOMENT	VALUE	UNIT	TORSIONAL MOMENT	VALUE	UNIT	
Max. Mx	2.863	kN-m	Corresponding Mxy	1.171	kN-m	*(Plate No. - 115 , L/C No. - 412)*
Min. Mx	-12.778	kN-m	Corresponding Mxy	3.370	kN-m	*(Plate No. - 24 , L/C No. - 404)*
Max. My	9.238	kN-m	Corresponding Mxy	0.629	kN-m	*(Plate No. - 114 , L/C No. - 412)*
Min. My	-19.510	kN-m	Corresponding Mxy	1.810	kN-m	*(Plate No. - 25 , L/C No. - 404)*

FOR TOP FACE REINFORCEMRNT :

Mx1 =	Mx + abs(Mxy) =	2.863 + 1.171 =	4.034	kN-m
My1 =	My + abs(Mxy) =	9.238 + 0.629 =	9.867	kN-m
Mx2 =	Mx+abs(Mxy^2/My) =	2.863 + 0.148 =	3.011	kN-m
My2 =	My+abs(Mxy^2/Mx) =	9.238 + 0.078 =	9.316	kN-m



DESIGN MOMENT CALCULATION FOR TOP FACE REINFORCEMENT :

Mxd =	4.034	kN-m
Myd =	9.867	kN-m

ALONG LOCAL X - DIRECTION :

Mu =	4.034	kN-m					
Φ*Kn =	Mu/bd^2				deff. =	319	mm
Mu/bd^2 =	4.034X10^6/(1000X319^2) =	0.04	N/mm^2	<	6.9	N/mm^2	
Pt =	0.0001	Reqd. Ast =	0.0001X1000X319/100 =	0.32	cm^2		
Provided	12 mm @	200 C/C	(Provide Ast =	5.65	$)cm^2$		
			Total Provide Ast =	5.65	cm^2	O.K.	
			Total Provide Pt =	0.0018			

ALONG LOCAL Y - DIRECTION :

Mu =	9.867	kN-m					
Φ*Kn =	Mu/bd^2				deff. =	307	mm
Mu/bd^2 =	9.867X10^6/(1000X307^2) =	0.097	N/mm^2	<	6.9	N/mm^2	
Pt =	0.0003	Reqd. Ast =	0.0003X1000X319/100 =	0.92	cm^2		
Provided	12 mm @	200 C/C	(Provide Ast =	5.65	$)cm^2$		
			Total Provide Ast =	5.65	cm^2	O.K.	
			Total Provide Pt =	0.0018			

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01	Outotec / L&T Revision No.:	0	1	2
Date	16-04-2020	13-05-2020	31.10.2023	Date	16-04-2020	13-05-2020	31.10.2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT	Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	APR	APR	APR	Prepared By	APR	APR	APR
Checked	SBY	SBY	SBY	Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI	Approved	PRI	PRI	PRI

FOR BOTTOM FACE REINFORCEMRNT :

Mx1 = Mx - abs(Mxy) = -12.778 - 3.37 = -16.148 kN-m
My1 = My - abs(Mxy) = -19.51 - 1.81 = -21.32 kN-m
Mx2 = Mx-abs(Mxy^2/My) = -12.778 - 0.582 = -13.36 kN-m
My2 = My-abs(Mxy^2/Mx) = -19.51 - 0.256 = -19.254 kN-m

DESIGN MOMENT CALCULATION FOR BOTTOM FACE REINFORCEMENT :

Mxd = -16.148 kN-m
Myd = -21.32 kN-m

ALONG LOCAL X - DIRECTION :

Mu = 16.148 kN-m
Φ*Kn = Mu/bd^2 deff. = 317 mm
Mu/bd^2 = 16.148X10^6/(1000X317^2) = 0.161 N/mm^2 < 6.9 N/mm^2
Pt = 0.0004 Reqd. Ast = 0.0004X1000X317/100 = 1.27 cm^2
Provided 16 mm @ 200 C/C (Provide Ast = 10.05)cm^2

Total Provide Ast = 10.05 cm^2 O.K.
Total Provide Pt = 0.0032

ALONG LOCAL Y - DIRECTION :

Mu = 21.32 kN-m
Φ*Kn = Mu/bd^2 deff. = 301 mm
Mu/bd^2 = 21.32X10^6/(1000X301^2) = 0.235 N/mm^2 < 6.9 N/mm^2
Pt = 0.0006 Reqd. Ast = 0.0006X1000X301/100 = 1.81 cm^2
Provided 16 mm @ 200 C/C (Provide Ast = 10.05)cm^2

Total Provide Ast = 10.05 cm^2 O.K.
Total Provide Pt = 0.0033

SHEAR CHECK

(Staad Group : SHEAR)
(FROM FACTORED LOAD COMBINATION)

ALONG LOCAL X - DIRECTION :

SQx = -0.029 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 166 , L/C No. - 404)*
Vu = 9.19 kN

Provided Pt = 0.0032
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0032X9.193X10^3X317/16.148/10^6)X1000X317/10^3
= 303.176 kN
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0032)X1000X317/10^3
= 317.308 kN
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X5.92X1000X317/10^3
= 543.865 kN
HENCE, Vc = 303.176 kN
Φt*Vc = 0.75X303.176 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 227.382 kN > 9.193T , Hence O.K.

ALONG LOCAL Y - DIRECTION :

SQy = 0.036 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 294 , L/C No. - 403)*
Vu = 10.836 kN

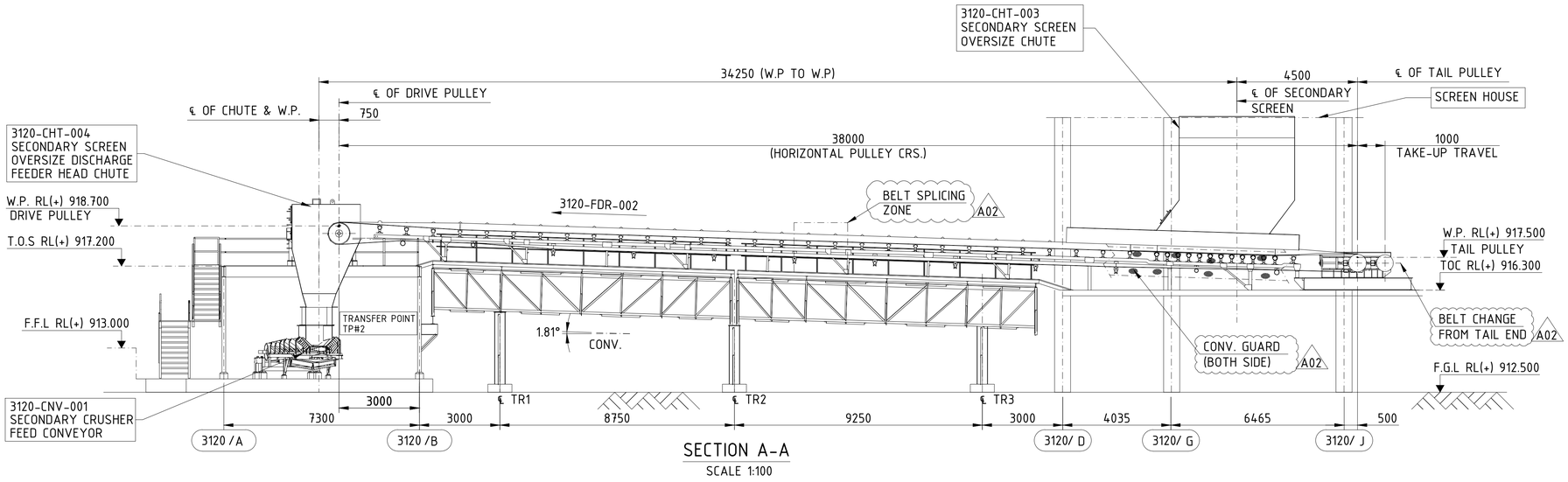
Pt provided = 0.0033
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0033X10.836X10^3X301/21.32/10^3)X1000X301/10^3
= 287.502 kN
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0033)X1000X301/10^3
= 301.805 KN
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X5.92X1000X301/10^3
= 516.415 kN
HENCE, Vc = 287.502 kN
Φt*Vc = 0.75X287.502 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 215.627 kN > 10.836T , Hence O.K.

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

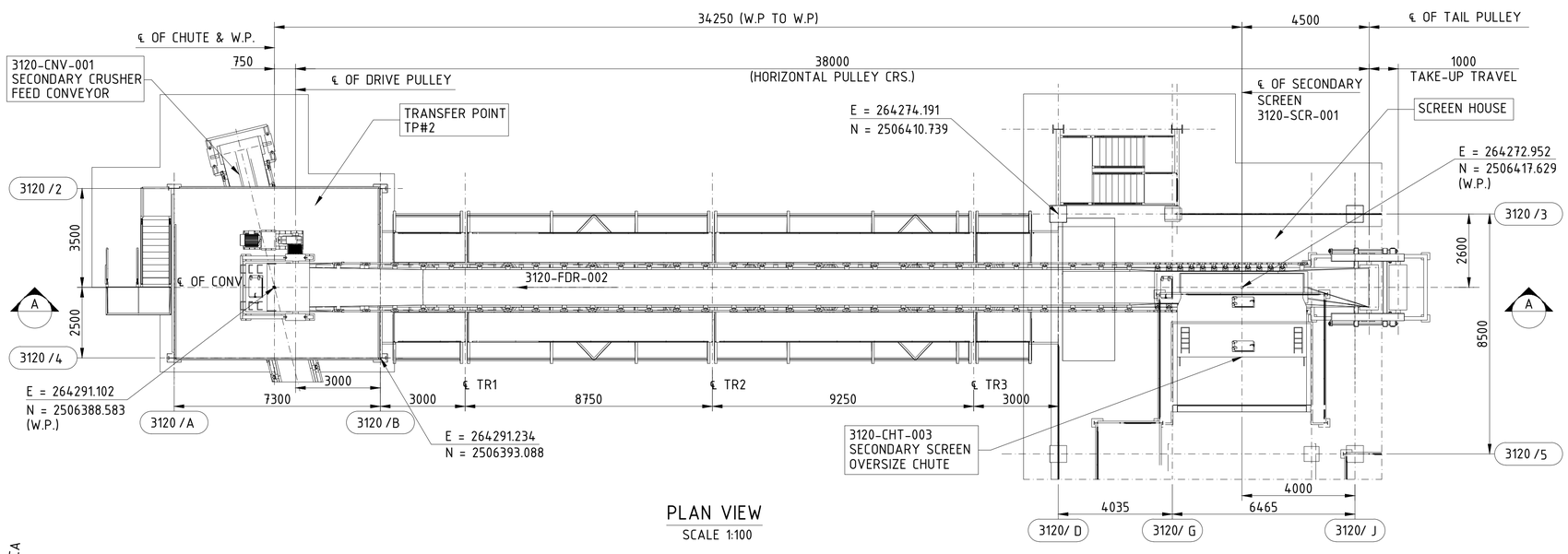
Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2			
Date	16-04-2020	13-05-2020	31.10.2023				Date	16-04-2020	13-05-2020	31.10.2023			
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT			
Prepared By	APR	APR	APR				Prepared By	APR	APR	APR			
Checked	SBY	SBY	SBY				Checked	SBY	SBY	SBY			
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI			

ATTACHMENT 1 **MECHANICAL GENERAL ARRANGEMENT DRAWINGS**

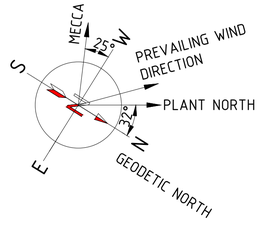
DRAWING NUMBER MD-216-3120-EG-ME-LPX-1003 REV A02



SECTION A-A
SCALE 1:100



PLAN VIEW
SCALE 1:100

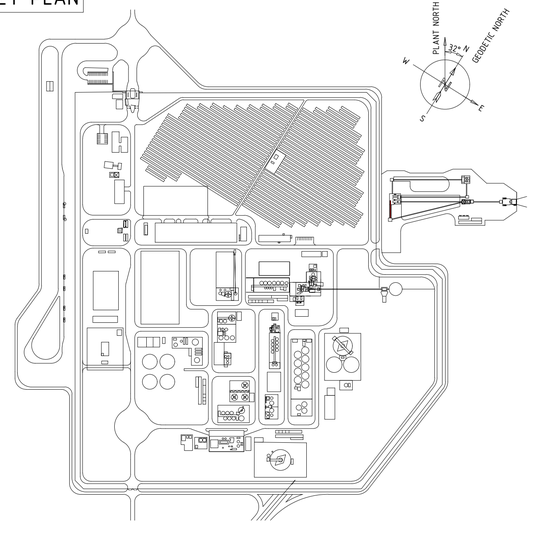


ABBREVIATION			
CONV.:	CONVEYOR	CRS.:	CENTERS
TBA:	TO BE ANNOUNCED	DS:	DUST SUPPRESSION
N.A.:	NOT APPLICABLE	VGTU.:	VERTICAL GRAVITY TAKE-UP
GP:	GROUND PORTAL	MKD:	MARKED
SARII:	SELF ALIGNING RETURN TRAINER INVERTED IDLER	DIA:	DIAMETER
		SYN:	SYNCHRONOUS
T.P.:	TANGENT POINT	thk.:	THICKNESS
I.P.:	INTERSECTION POINT	RCC:	REINFORCED CEMENT CONCRETE
W.P.:	WORKING POINT	B/W:	BELT WIDTH
STD.:	STANDARD	MAX:	MAXIMUM
STR.:	STRINGER	deg.:	DEGREE
SEC:	SECTION	U/S.:	UNDERSIDE
℄:	CENTER LINE	B.O.M.R:	BOTTOM OF MONORAIL
DN:	DOWNWARD	T.O.W.W.:	TOP OF WALKWAY
UP:	UPWARD	TR:	TRESTLE
TP:	TRANSFER POINT	TYP.:	TYPICAL
BRG.:	BEARING	OPNG.:	OPENING

NOTE:
1. ALL DIMENSIONS ARE IN MILLIMETER, CO-ORDINATES AND ELEVATION ARE IN METER, UNLESS OTHERWISE STATED.
2. FINISHED FLOOR LEVEL (F.F.L) FOR CRUSHING AREA = RL(+) 913.000 m
3. FINISHED GROUND LEVEL (F.G.L) FOR CRUSHING AREA = RL(+) 912.500 m
4. F.F.L, F.G.L, T.O.C, = FINISH FLOOR LEVEL, FINISH GROUND LEVEL, TOP OF CONCRETE, T.O.S, B.O.S, B.O.B= TOP OF STEEL, BOTTOM OF STEEL, BOTTOM OF BASE PLATE
5. ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE ARE SHOWN TENTATIVE.

CONCRETE FLOOR
CHECKERED PLATE
GRATING
BRICK WALL
CONCRETE WALL

KEY PLAN



DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		A02	04/04/2020	ISSUED FOR DETAIL DESIGN & REVISED AS PER EMPLOYER	AFO	BRD	BD	BGR	
				COMMENTS DT. 27.03.2020 & REVISED AS MARKED A02					
MD-216-3120-EG-ME-ELV-1005	FEEDER 3120-FDR-002-ELEVATION	A01	04/03/2020	ISSUED FOR APPROVAL	AFO	BRD	BD	BGR	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

LARSEN & TOUBRO

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MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

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LARSEN & TOUBRO DRAWING NUMBER		REVISION
O19045_100_H_G_16117_001		1
OUTOTEC DRAWING NUMBER	DOCUMENT ID	REVISION
900707__GAU01__ZC01__MTL01__31266	OU100670526	1

DRAWN	AFO	28/02/2020
CHECKED	BRD	02/03/2020
ENGINEER	BD	25/02/2020
APPROVED	BGR	04/03/2020
EMPLOYER		
SCALE	1:100	A1

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
SECONDARY CRUSHER
FEEDER 3120-FDR-002 - PLAN VIEW

DRAWING NUMBER: MD-216-3120-EG-ME-LPX-1003 REV A02

TABLE-1 : LOAD FOR CONVEYOR SHORT POST

FOR CONV. 3120-FDR-002

POINT	VERTICAL LOAD IN "t"		HORIZONTAL LOAD IN "t"	
	DEAD LOAD	LIVE LOAD	ALONG	ACROSS
F1	(+)0.60	(+)0.65 A02	(±)0.15	(±)0.08

SIGN CONVENTION

UPWARD (-)
DOWNWARD (+)
TOWARDS DRIVE PULLEY=(+)
TOWARDS TAIL PULLEY=(-)
REVERSIBLE LOAD=(±)

Note: Wind & seismic load to be considered during civil & structural design

CONVEYOR CHARACTERISTICS

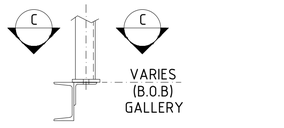
	PARTICULARS	SPECIFICATION	QUANTITY
MISCELLANEOUS EQUIPMENT	TAKE-UP / TRAVEL	SCREW / 1000 mm	
	TAKE-UP LOAD	N.A.	
	SCRAPER AT DISCHARGE PULLEY	EXTERNAL / MAIN & PRE CLEANER	1 NO. + 1 NO.
	SCRAPER AT TAKE-UP	N.A.	
	SCRAPER AT TAIL PULLEY	INTERNAL V-PLOUGH	1 NO.
	BELT SWAY SWITCH (ZS)	@ 50 m INTERVAL	2 PAIRS
	PULL CORD SWITCH (HS)	@ 30 m INTERVAL	2 PAIRS
	ZERO SPEED SWITCH (SS)	AT TAIL PULLEY	1 NO.
	BELT RIP SWITCH (YS)	AFTER FEEDING ZONE	1 NO.
	DECK PLATE	3.15 mm thk. CONTINUOUS	1 LOT.
	BELT HOOD	GALVANIZED CORRUGATED SHEET	1 LOT.

ABBREVIATION

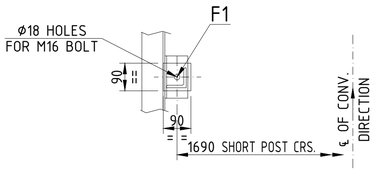
CONV.:	CONVEYOR	CRS.:	CENTERS
TBA:	TO BE ANNOUNCED	DS:	DUST SUPPRESSION
N.A.:	NOT APPLICABLE	VGTU.:	VERTICAL GRAVITY TAKE-UP
GP:	GROUND PORTAL	MKD:	MARKED
SARII:	SELF ALIGNING RETURN TRAINER INVERTED IDLER	DIA:	DIAMETER
		SYN:	SYNCHRONOUS
T.P.:	TANGENT POINT	thk.:	THICKNESS
I.P.:	INTERSECTION POINT	RCC:	REINFORCED CEMENT CONCRETE
W.P.:	WORKING POINT	B/W:	BELT WIDTH
STD.:	STANDARD	MAX:	MAXIMUM
STR.:	STRINGER	deg.:	DEGREE
SEC:	SECTION	U/S.:	UNDERSIDE
℄:	CENTER LINE	B.O.M.R:	BOTTOM OF MONORAIL
DN:	DOWNWARD	T.O.W.W.:	TOP OF WALKWAY
UP:	UPWARD	TR:	TRESTLE
TP:	TRANSFER POINT	TYP.:	TYPICAL
BRG.:	BEARING	OPNG.:	OPENING

CONVEYOR CHARACTERISTICS

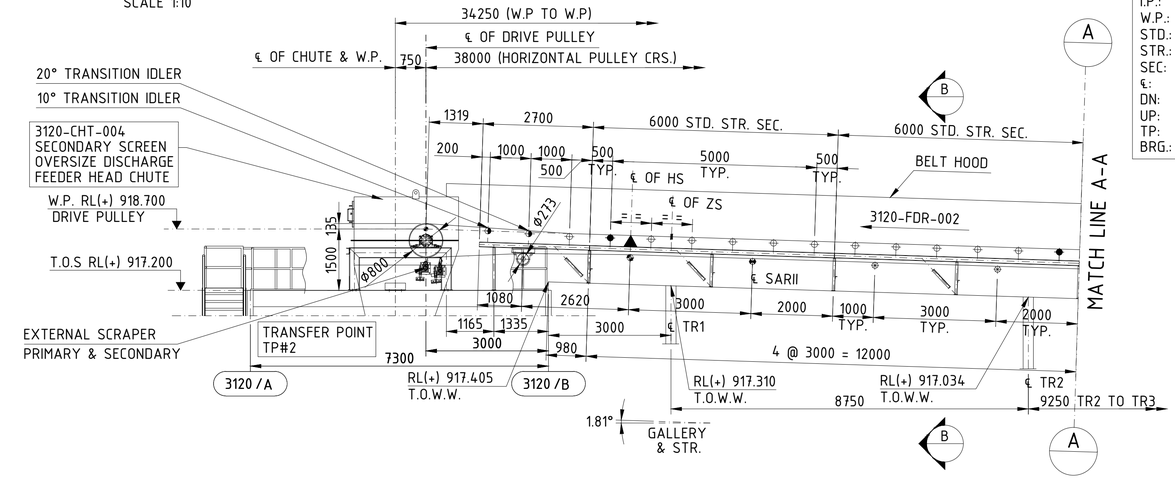
		PARTICULARS	SPECIFICATION	QUANTITY
CONVEYOR		EQUIPMENT TAG NO.	3120-FDR-002	1 NO.
		EQUIPMENT DESCRIPTION	SECONDARY SCREEN OVERSIZE DISCHARGE FEEDER	
		LOCATION	SCREEN HOUSE TO TRANSFER POINT TP#2	
		CAPACITY RATED / DESIGN (t/h)	772 / 850	
		BELT WIDTH (mm)	1400	
		BELT SPEED (m/sec)	1.537	
		TRIPPER FIXED / MOVABLE	N.A.	
		CONVEYOR LENGTH / LIFT (mm)	38000 / 1200	
MATERIAL		MATERIAL	GOLD ORE : FRESH ORE / OXIDE ORE	
		BULK DENSITY (t/m³)	1.6	
		MAX. LUMP SIZE (mm)	(-)340	
		MOISTURE CONTENT (%)	2	
		SURCHARGE ANGLE (deg.)	20	
BELTING		MAX. RUNNING TENSION (T1 kN)	58.229	
		SLACK SIDE TENSION (T2 kN)	48.986	
		TAIL TENSION (T0 kN)	52.136	
		BELT DESIGNATION	EP-1000 / 4 (HEAVY DUTY)	
		TOP / BOTTOM COVER THICKNESS (mm)	8 / 3	
		COVER GRADE / FABRIC	DIN-X / EP	
PULLEYS		DISCHARGE WITH DRIVE (MKD. x DIA.)	D-2 x 800 mm	1 NO.
		DISCHARGE WITHOUT DRIVE (MKD. x DIA.)	N.A.	
		SNUB (MKD. x DIA.)	N.A.	
		TAIL (MKD. x DIA.)	ND-3 x 630 mm	1 NO.
		TAKE-UP (MKD. x DIA.)	N.A.	
		DEFLECTOR (MKD. x DIA.)	DR-2 x 273 mm	1 NO.
		TRIPPER DISCHARGE (MKD. x DIA.)	N.A.	
		TRIPPER BEND (MKD. x DIA.)	N.A.	
		BEND (MKD. x DIA.)	N.A.	
		LAGGING (mm)	DRIVE: 16thk CERAMIC	
			NON-DRIVE: 16thk PLAIN RUBBER	
BEARING/SLEEVE PLUMMER BLOCK		DISCHARGE WITH DRIVE	TBA	2 SETS
		DISCHARGE WITHOUT DRIVE	N.A.	
		SNUB	TBA	
		TAIL	TBA	2 SETS
		TAKE-UP	N.A.	
		DEFLECTOR	TBA	2 SETS
		BEND	TBA	
		TRIPPER DISCHARGE	N.A.	
		TRIPPER BEND	N.A.	
IDLER		TROUGHING (DIA. x deg.)	152.4 x 35°	21 NOS.
		TROUGHING RETRACTABLE (DIA. x deg.)	152.4 x 45°	9 NOS.
		TROUGHING TRAINER (DIA. x deg.)	152.4 x 35°	3 NOS.
		RETURN (DIA. x deg.)	152.4 x 10°	7 NOS.
		SELF ALIGNING RETURN TRAINER INVERTED IDLER (DIA. x deg.)	152.4 x 10°	3 NOS.
		IMPACT RETRACTABLE (DIA. x deg.)	190 x 45°	12 NOS.
		TRANSITION (ADJUSTABLE) (DIA. x deg.)	152.4 x 10°/20°/30°	(1+1+1) NOS.
		RETURN SELF CLEANING (DIA. x deg.)	190 x 0°	1 NO.
		NON MAGNETIC TROUGHING (DIA. x deg.)	152.4 x 35°	
		WEIGHING IDLER (DIA. x deg.)	152.4 x 35°	
		SPECIAL MACHINED IDLER (DIA. x deg.)	152.4 x 35°	
		RETURN FLAT IDLER (DIA. x deg.)	152.4 x 0°	2 NOS.
DRIVE	MOTOR	RATING kW/rpm (SYN)	45 / 1800	1 NO.
		FRAME SIZE	TBA	
		TYPE	SQUIRREL CAGE	
		SUPPLY	400V, 60Hz	
	REDUCER	TYPE	HELICAL	1 NO.
		MODEL NO./COOLING/RATIO	TBA	
		RATIO	50	
		HOLD BACK	YES	
	BRAKE	TYPE	TBA	
		MODEL NO.	TBA	
		DRAM DIAMETER	TBA	
	COUPLING LOW SPEED	TYPE	RESILIENT COUPLING	1 NO.
	COUPLING HIGH SPEED	MODEL NO.	TBA	
		TYPE	GEAR	1 NO.
		MODEL NO.	TBA	

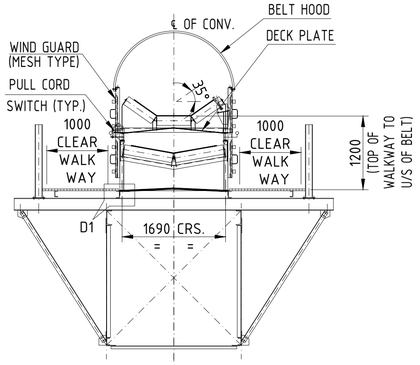


DETAIL-D1
SCALE 1:10

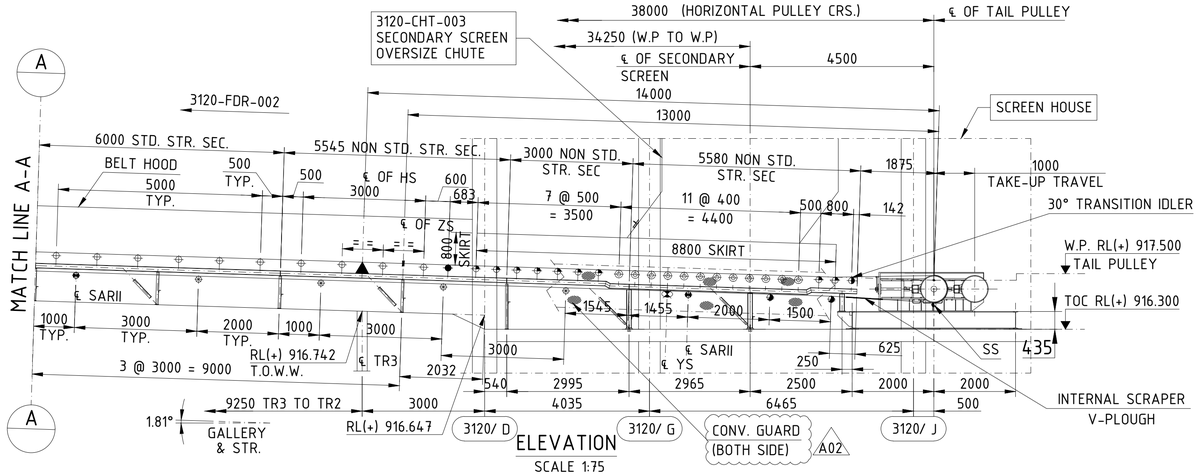


VIEW C-C
SCALE 1:10





VIEW B-B
TYPICAL CONVEYOR SECTION ON GALLERY
(1400 B/W)
SCALE 1:50



ELEVATION
SCALE 1:75

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
A02	04/04/2020	ISSUED FOR DETAIL DESIGN & REVISED AS PER EMPLOYER COMMENTS DT. 27.03.2020 & REVISED AS MARKED A02	AFO	BRD	BD	BGR	
A01	04/03/2020	ISSUED FOR APPROVAL	AFO	BRD	BD	BGR	

DOCUMENT NUMBER	REFERENCE DOCUMENTS
MD-216-3120-EG-ME-LPX-1003	FEEDER 3120-FDR-002-PLAN VIEW

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LARSEN & TOUBRO DRAWING NUMBER: 019045_100_H_G_16117_002 — REVISION: 1
OUTOTEC DRAWING NUMBER: 900707__GAU01__ZC01__MTL01__31233 — DOCUMENT ID: OU100669056 — REVISION: 1

DRAWN	AFO	28/02/2020
CHECKED	BRD	02/03/2020
ENGINEER	BD	25/02/2020
APPROVED	BGR	04/03/2020
EMPLOYER		
SCALE	1:75, 1:50, 1:10	A1

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
SECONDARY CRUSHING
FEEDER 3120-FDR-002 - ELEVATION

DRAWING NUMBER: MD-216-3120-EG-ME-ELV-1005 — REV: A02

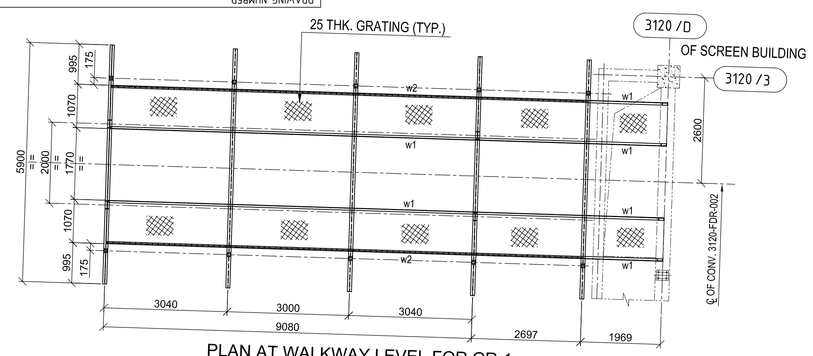
Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
Date	16-04-2020	13-05-2020	31.10.2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	APR	APR	APR
Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI

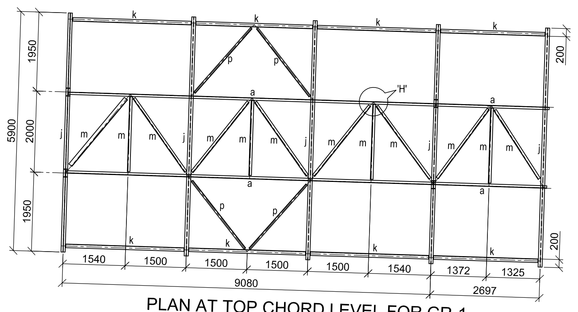
Outotec / L&T Revision No.:	0	1	2
Date	16-04-2020	13-05-2020	31.10.2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	APR	APR	APR
Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI

ATTACHMENT 2 **STRUCTURAL GENERAL ARRANGEMENT DRAWING CONTAINING BASE PLATE LAYOUT**

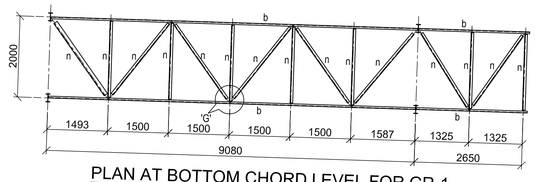
DRAWING NUMBER: MD - 216 - 3120 - EG - ST - GAD - 1022 | REV: A01



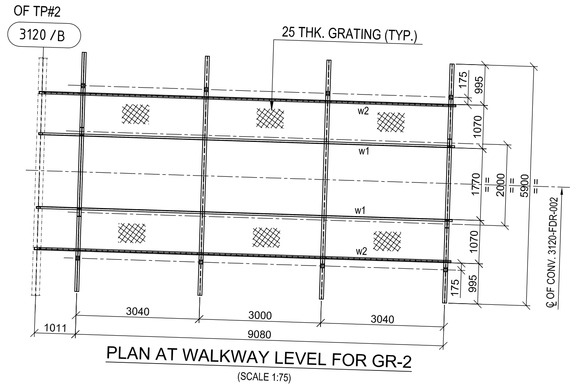
PLAN AT WALKWAY LEVEL FOR GR-1
(SCALE 1:75)



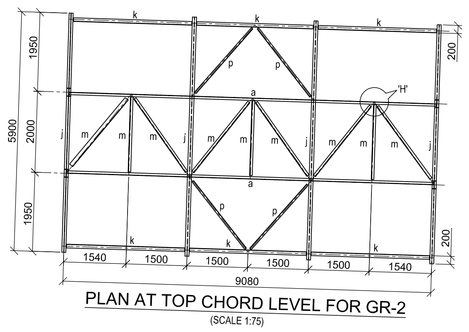
PLAN AT TOP CHORD LEVEL FOR GR-1
(SCALE 1:75)



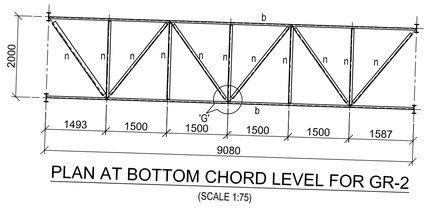
PLAN AT BOTTOM CHORD LEVEL FOR GR-1
(SCALE 1:75)



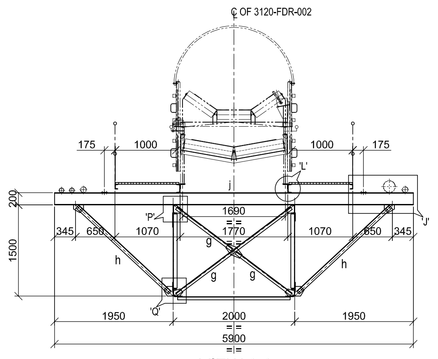
PLAN AT WALKWAY LEVEL FOR GR-2
(SCALE 1:75)



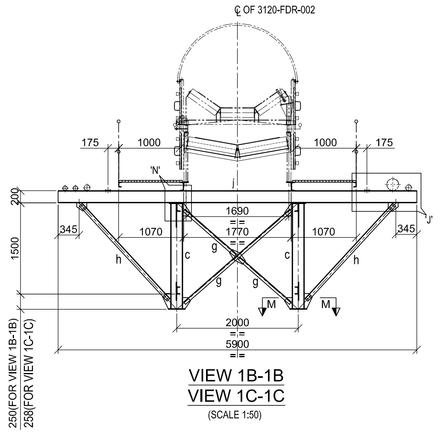
PLAN AT TOP CHORD LEVEL FOR GR-2
(SCALE 1:75)



PLAN AT BOTTOM CHORD LEVEL FOR GR-2
(SCALE 1:75)



VIEW 1-1
(SCALE 1:50)

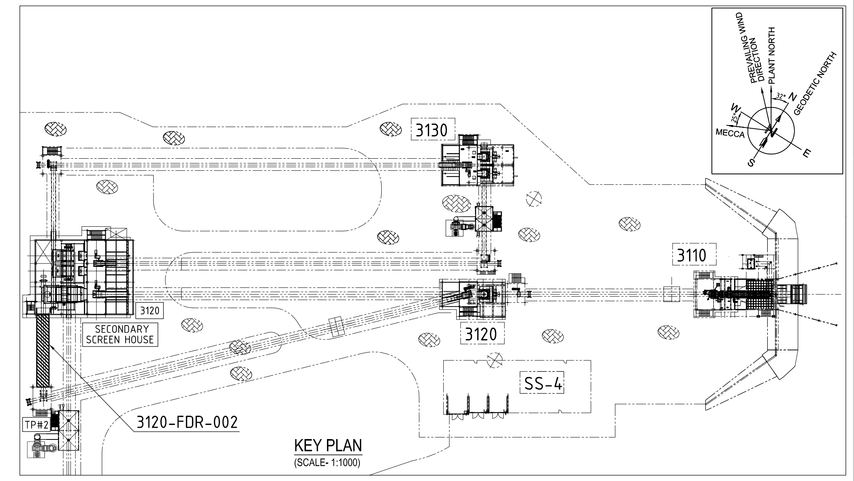


VIEW 1B-1B
VIEW 1C-1C
(SCALE 1:50)

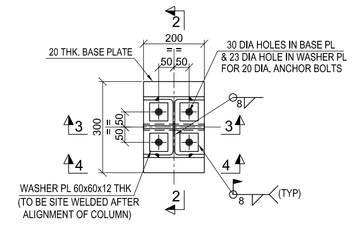
SCHEDULE OF MEMBERS FOR GALLERY & TRESTLE

MEMBER MARKED	MEMBER	SECTON	FACTORED AXIAL FORCE(KN)	FACTORED SHEAR FORCE(KN)	REMARKS
a	UPN 140	[	130	-	GUSSET PLATE 10 THK.
b	UPN 100	[	130	-	
c	IPE 200	I	-	-	
d	L 65X65X6	L	40	-	
e	L 90X90X8	L	90	-	
f	L 50X50X6	L	80	-	
g	L 50X50X6	L	20	-	GUSSET PLATE 8 THK.
h	L 65X65X6	L	20	-	
j	IPE 200	I	-	-	
k	L 90X90X8	L	-	-	
m	L 65X65X6	L	15	-	
n	L 65X65X6	L	15	-	
p	L 65X65X6	L	-	-	
s	UPN 100	[	-	-	
t	L 75X75X8	L	-	-	
u	L 50X50X6	L	-	-	
u1	2L 50X50X6	⊓	75	-	
w1	UPN 140	[	-	-	
w2	UPN 100	[	-	-	
C1	1. PL. 130X12 2. PL. 236X8	I (1, 2)	-	-	

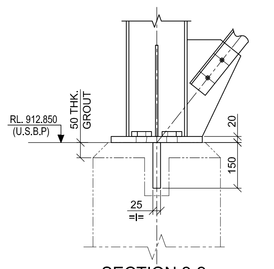
GRADE OF MATERIAL IS S275 JR.
FOR DEVELOPING CONNECTIONS MEMBER END FORCES MENTIONED IN SCHEDULE OF MEMBERS SHALL BE FOLLOWED.
MINIMUM 6mm FILLET WELD OF 150 mm LENGTH SHALL BE PROVIDED FOR WELDED CONNECTION, FOR BOLTED CONNECTION, MINIMUM 2 Nos. M20 BOLTS SHALL BE PROVIDED.



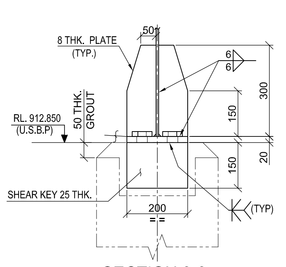
KEY PLAN
(SCALE- 1:1000)



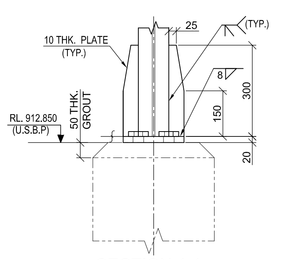
BASE PLATE DETAIL FOR TR1
(SCALE - 1:10)



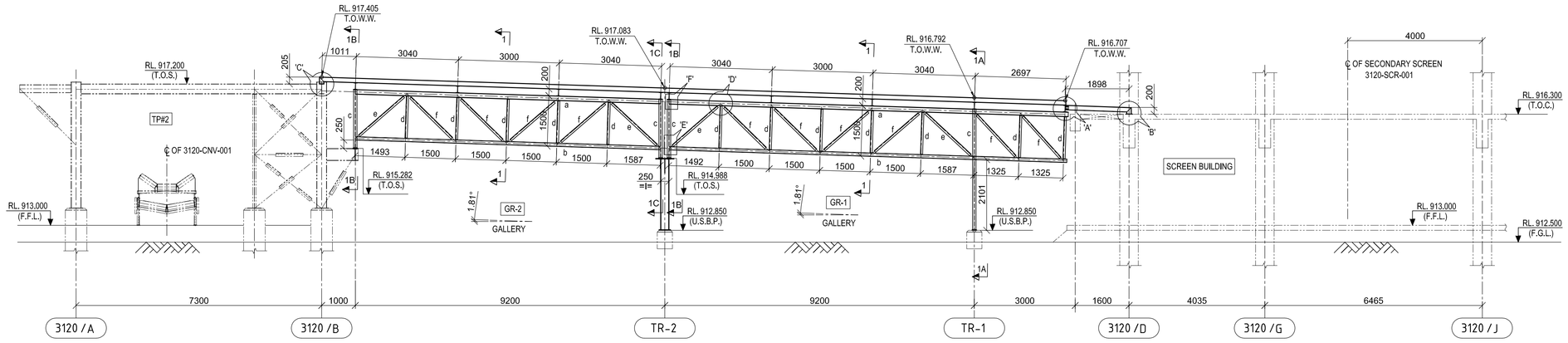
SECTION 2-2
(SCALE - 1:10)



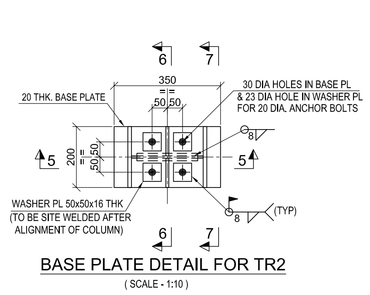
SECTION 3-3
(SCALE - 1:10)



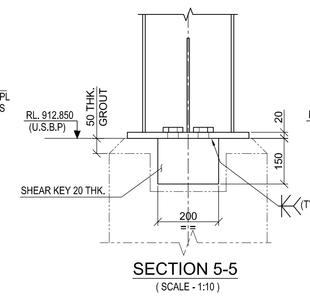
SECTION 4-4
(SCALE - 1:10)



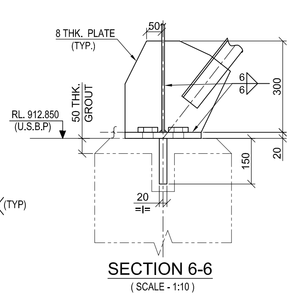
VIEW A-A
ELEVATION OF CONVEYOR MKD. 3120-FDR-002
(SCALE 1:75)



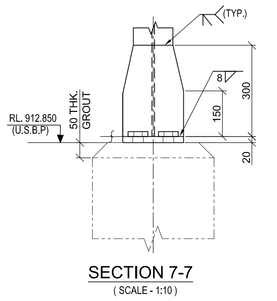
BASE PLATE DETAIL FOR TR2
(SCALE - 1:10)



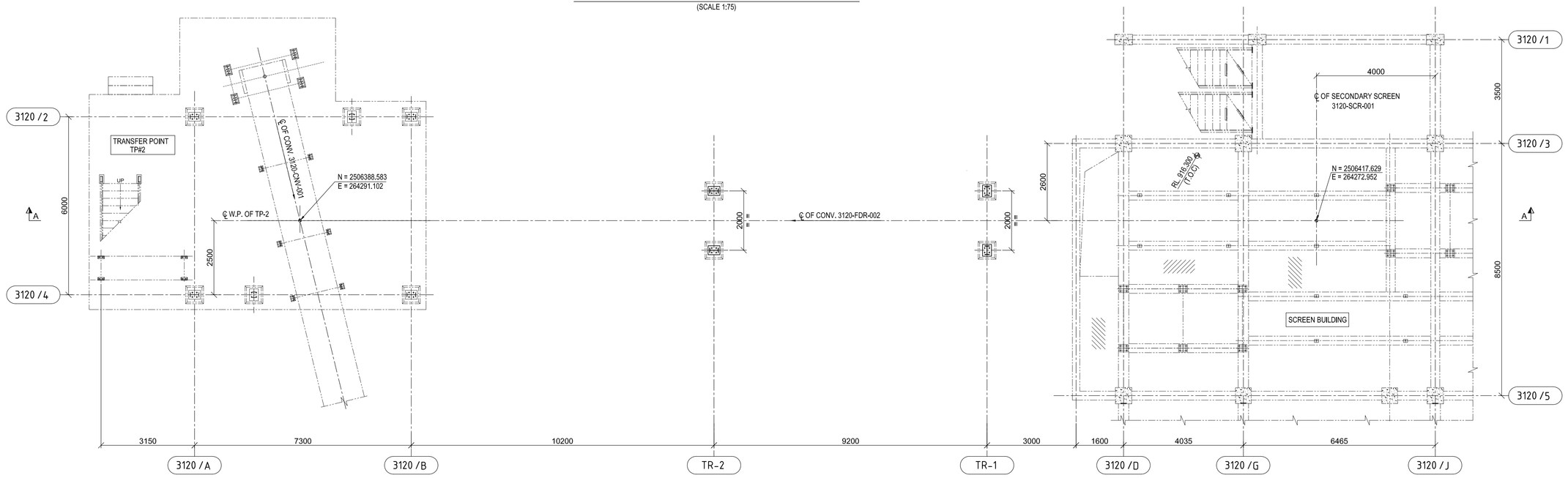
SECTION 5-5
(SCALE - 1:10)



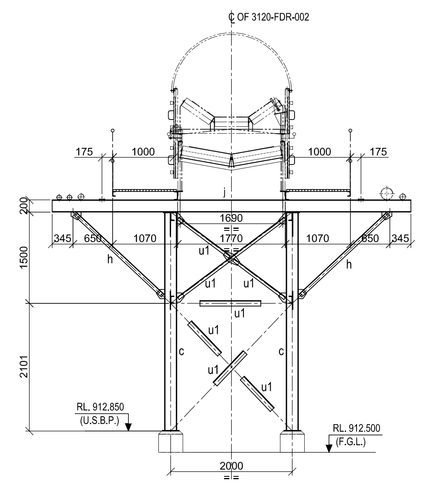
SECTION 6-6
(SCALE - 1:10)



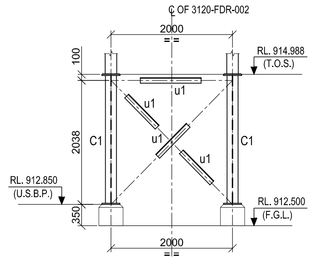
SECTION 7-7
(SCALE - 1:10)



PLAN AT BASE PLATE LEVEL OF CONVEYOR MKD. 3120-FDR-002
(SCALE 1:75)



VIEW 1A-1A
ELEVATION OF TRESTLE MKD. TR1
(SCALE 1:50)



ELEVATION OF TRESTLE MKD. TR2
(SCALE 1:50)

LEGENDS :-

R.L. - REDUCED LEVEL
℄ - CENTRE LINE
T.O.S. - TOP OF STEEL.
F.G.L. - FINISHED GROUND LEVEL.
F.F.L. - FINISHED FLOOR LEVEL.
TYP. - TYPICAL
MKD. - MARKED.
THK. - THICK
F/S - FAR SIDE
N/S - NEAR SIDE
W.P.-WORKING POINT
U.S.B.P. - UNDER SIDE OF BASE PLATE
U.N.O. - UNLESS NOTED OTHERWISE
T.O.W.W. - TOP OF WALKWAY
REF. - REFER
DRG. NO. - DRAWING NUMBER
STRL. - STRUCTURAL

NOTES :-

1. ALL DIMENSIONS ARE IN MILLIMETRES AND LEVELS AND CO-ORDINATES ARE IN METRES. UNLESS OTHERWISE STATED.
2. GRADE OF STEEL USED S275R AS PER EN10025.
3. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH DRAWING MD - 216 - 3120 -EG -ST - GAD - 1023.
4. ALL STRUCTURAL WORK SHALL CONFORM TO CIVIL STRUCTURAL DESIGN CRITERIA NO MD - 216 - 1000 -EG -ST - CRT - 1001

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3120-EG-ME-ELV-1005	FEEDER 3120-FDR-002 - ELEVATION								
MD-216-3120-EG-ME-LPX-1003	FEEDER 3120-FDR-002 - PLAN VIEW	A01	15/04/2020	ISSUED FOR REFERENCE	HSK	SBY	APR	PRI	

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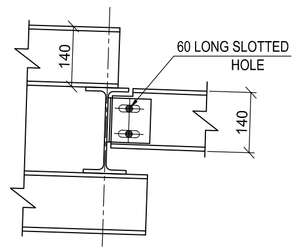
MA'ADEN معادن
Saudi Arabian Mining Company شركة معادن العربية السعودية

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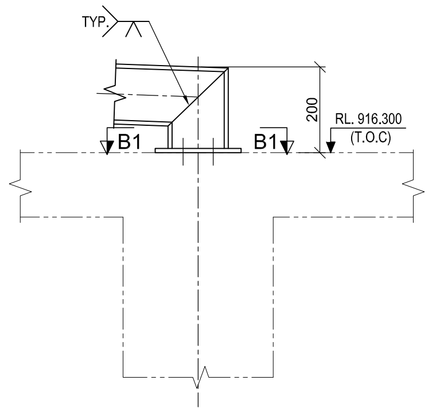
LARSEN & TOUBRO DRAWING NUMBER: 019045_100_S_G_26013_032 | REVISION: 0
OUTOTEC DRAWING NUMBER: 900707 _ GAU01 _ US01 _ MTL01 _ 32136 | DOCUMENT ID: OU100680976 | REVISION: 0

DRAWN	HSK	13/04/2020	MANSOURAH - MASSARAH GOLD PROJECT
CHECKED	SBY	14/04/2020	GENERAL ARRANGEMENT DRAWING
ENGINEER	APR	10/04/2020	STRL. GA OF CONVEYOR-3120-FDR-002,SHT 1
APPROVED	PRI	15/04/2020	PLAN, ELEVATIONS & DETAILS
EMPLOYER			
SCALE	1:100,50,20,10	A0	DRAWING NUMBER: MD - 216 - 3120 - EG - ST - GAD - 1022 | REV: A01

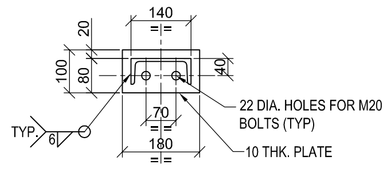
DRAWING NUMBER MD-216-3120-EG-ST-GAD-1023 REV A01



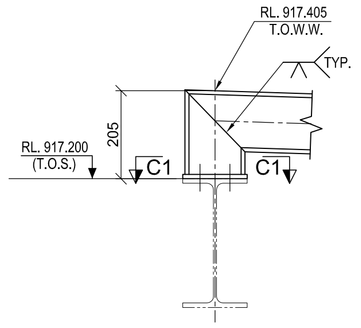
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(SCALE 1:10)



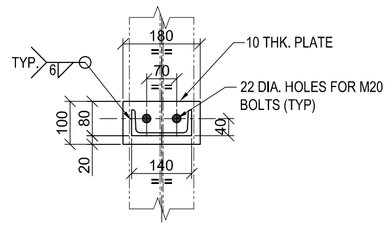
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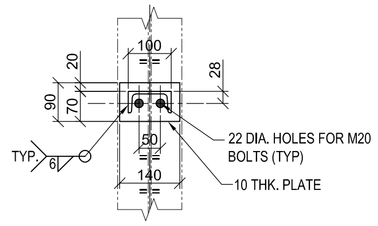
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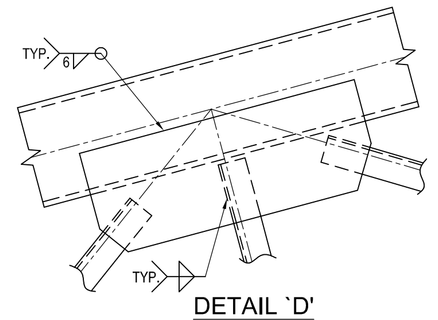
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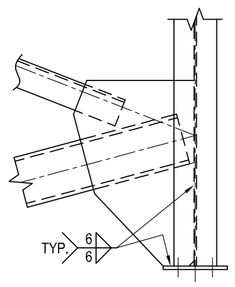
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(FOR W1)
(SCALE 1:10)



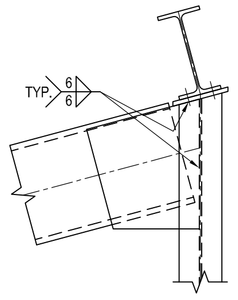
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(FOR W2)
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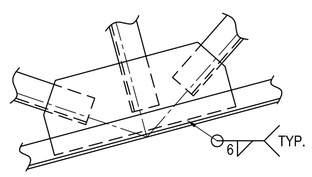
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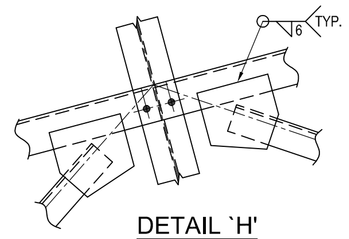
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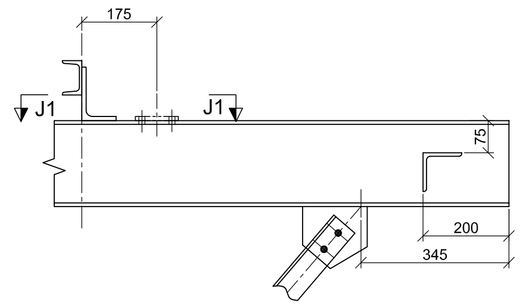
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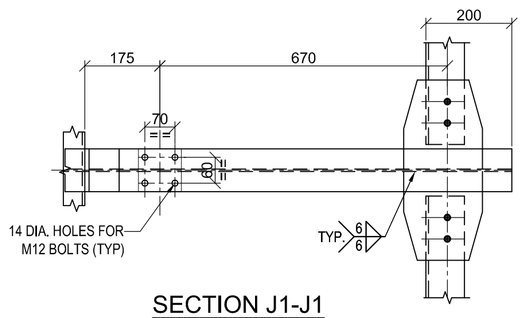
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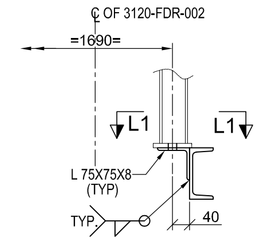
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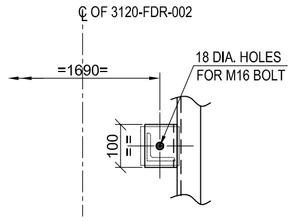
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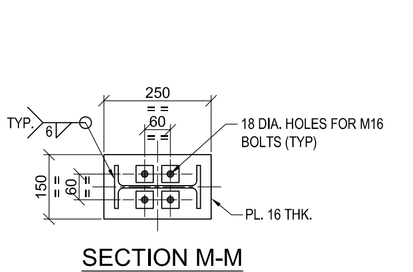
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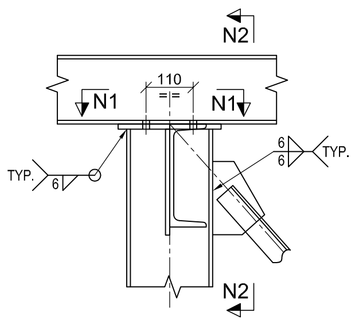
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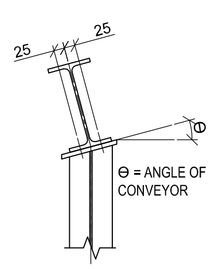
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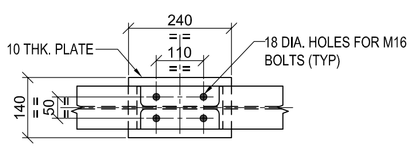
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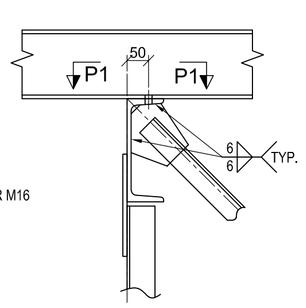
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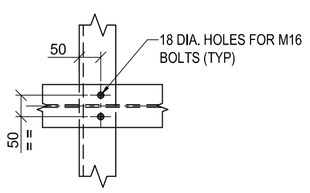
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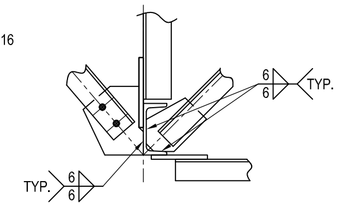
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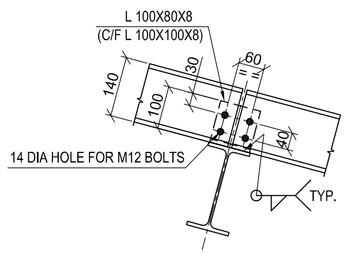
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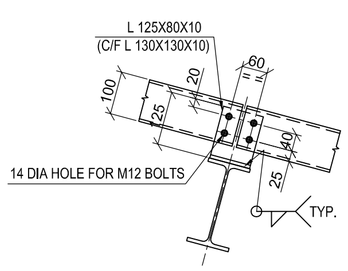
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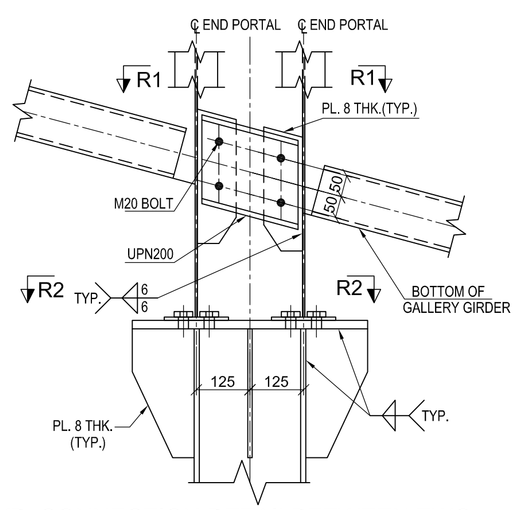
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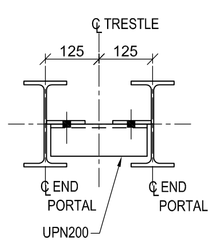
TYP. CONNECTION DETAIL FOR WALKWAY SUPPORTING CHANNEL FOR W1
(SCALE 1:10)



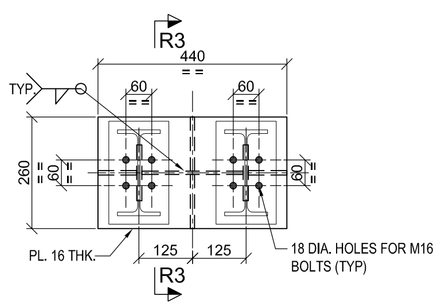
TYP. CONNECTION DETAIL FOR WALKWAY SUPPORTING CHANNEL FOR W2
(SCALE 1:10)



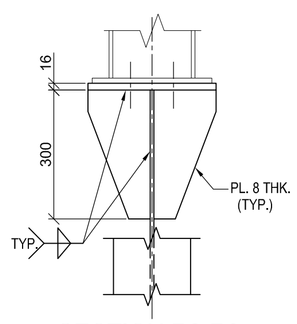
TYP. CONNECTION DETAILBETWEEN TWO ADJUSCENT END PORTAL
(REF. DRG. NO. 1022)
(SCALE 1:10)



SECTION R1-R1
(SCALE 1:10)



SECTION R2-R2
(SCALE 1:10)



SECTION R3-R3
(SCALE 1:10)

LEGENDS :-

R.L. - REDUCED LEVEL
℄ - CENTRE LINE
T.O.S. - TOP OF STEEL.
F.G.L. - FINISHED GROUND LEVEL.
F.F.L. - FINISHED FLOOR LEVEL.
TYP. - TYPICAL.
MKD. - MARKED.
THK. - THICK
F/S - FAR SIDE
N/S - NEAR SIDE
W.P.-WORKING POINT
U.S.B.P. - UNDER SIDE OF BASE PLATE
U.N.O. - UNLESS NOTED OTHERWISE
T.O.W.W. - TOP OF WALKWAY
REF. - REFER
DRG. NO. - DRAWING NUMBER
STRL. - STRUCTURAL

NOTES :-

1. ALL DIMENSIONS ARE IN MILLIMETRES AND LEVELS AND CO-ORDINATES ARE IN METRES. UNLESS OTHERWISE STATED.
2. GRADE OF STEEL USED S275R AS PER EN10025.
3. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH DRAWING MD - 216 - 3120 -EG -ST - GAD - 1022.
4. ALL STRUCTURAL WORK SHALL CONFORM TO CIVIL STRUCTURAL DESIGN CRITERIA NO MD - 216 - 1000 -EG -ST - CRT - 1001.

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3120-EG-ME-ELV-1005	FEEDER 3120-FDR-002 - ELEVATION								
MD-216-3120-EG-ME-LPX-1003	FEEDER 3120-FDR-002 - PLAN VIEW	A01	15/04/2020	ISSUED FOR REFERENCE	HSK	SBY	APR	PRI	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

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DRAWN	HSK	13/04/2020
CHECKED	SBY	14/04/2020
ENGINEER	APR	10/04/2020
APPROVED	PRI	15/04/2020
EMPLOYER		
SCALE	1:100,20,15	A1

LARSEN & TOUBRO DRAWING NUMBER: 019045_100_S_G_26013_033 — REVISION: 0
OUTOTEC DRAWING NUMBER: 900707__GAU01__US01__MTL01__32137 — DOCUMENT ID: OU100680977 — REVISION: 0

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
STRL. GA OF CONVEYOR-3120-FDR-002,SHT 2
PLAN, ELEVATIONS & DETAILS

DRAWING NUMBER: MD-216-3120-EG-ST-GAD-1023 — REV: A01

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2			
Date	16-04-2020	13-05-2020	31.10.2023				Date	16-04-2020	13-05-2020	31.10.2023			
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT			
Prepared By	APR	APR	APR				Prepared By	APR	APR	APR			
Checked	SBY	SBY	SBY				Checked	SBY	SBY	SBY			
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI			

ATTACHMENT 3 **STAAD INPUT & OUTPUT FILE OF FOUNDATION FOR TR-1**

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 15-Apr-20
JOB NAME MMGP-SAUDI ARABIA
JOB CLIENT MA'ADEN
JOB NO O19045/900707
JOB REV 0
JOB PART TRESTLE TR1 FOUNDATION
JOB REF FOUNDATIONS FOR CONVEYOR 3120-FDR-002
ENGINEER NAME APR
CHECKER NAME SBY
APPROVED NAME PRI
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
1 0 -3 0; 2 0 -3 2; 3 0 -1 0; 4 0 -1 2; 5 0 -0.65 0; 6 0 -0.65 2;
14 -0.75 -3 2.75; 17 -0.5 -3 2.75; 19 -0.25 -3 2.75; 20 0 -3 2.75;
22 0.25 -3 2.75; 24 0.5 -3 2.75; 26 0.75 -3 2.75; 28 -0.75 -3 2.5;
30 -0.5 -3 2.5; 31 -0.25 -3 2.5; 32 0 -3 2.5; 33 0.25 -3 2.5; 34 0.5 -3 2.5;
35 0.75 -3 2.5; 37 -0.75 -3 2.25; 39 -0.5 -3 2.25; 40 -0.25 -3 2.25;
41 0 -3 2.25; 42 0.25 -3 2.25; 43 0.5 -3 2.25; 44 0.75 -3 2.25; 46 -0.75 -3 2;
48 -0.5 -3 2; 49 -0.25 -3 2; 50 0.25 -3 2; 51 0.5 -3 2; 52 0.75 -3 2;
54 -0.75 -3 1.75; 56 -0.5 -3 1.75; 57 -0.25 -3 1.75; 58 0 -3 1.75;
59 0.25 -3 1.75; 60 0.5 -3 1.75; 61 0.75 -3 1.75; 63 -0.75 -3 1.5;
65 -0.5 -3 1.5; 66 -0.25 -3 1.5; 67 0 -3 1.5; 68 0.25 -3 1.5; 69 0.5 -3 1.5;
70 0.75 -3 1.5; 72 -0.75 -3 1.25; 74 -0.5 -3 1.25; 75 -0.25 -3 1.25;
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222 1.00033 -3 1.5; 224 1.00033 -3 1.25; 226 1.00033 -3 1; 228 1.00033 -3 0.75;
230 1.00033 -3 0.5; 232 1.00033 -3 0.25; 234 1.00033 -3 0;
236 1.00033 -3 -0.25; 238 1.00033 -3 -0.5; 240 1.00033 -3 -0.75;
242 1.00033 -3 -1; 247 1 -3 3; 260 -1.00033 -3 2.75; 261 -1.00033 -3 2.5;
263 -1.00033 -3 2.25; 265 -1.00033 -3 2; 267 -1.00033 -3 1.75; 269 -1 -3 1.5;
271 -1.00033 -3 1.25; 273 -1.00033 -3 1; 275 -1.00033 -3 0.75;
277 -1.00033 -3 0.5; 279 -1.00033 -3 0.25; 281 -1.00033 -3 0;
283 -1.00033 -3 -0.25; 285 -1.00033 -3 -0.5; 287 -1.00033 -3 -0.75;
289 -1.00033 -3 -1; 295 -1.00033 -3 3;
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MEMBER INCIDENCES
1 1 3; 2 3 5; 3 2 4; 4 4 6; 5 3 4;
ELEMENT INCIDENCES SHELL
15 14 17 30 28; 16 17 19 31 30; 17 19 20 32 31; 18 20 22 33 32; 19 22 24 34 33;
20 24 26 35 34; 23 28 30 39 37; 24 30 31 40 39; 25 31 32 41 40; 26 32 33 42 41;
27 33 34 43 42; 28 34 35 44 43; 31 37 39 48 46; 32 39 40 49 48; 33 40 41 2 49;
34 41 42 50 2; 35 42 43 51 50; 36 43 44 52 51; 39 46 48 56 54; 40 48 49 57 56;
41 49 2 58 57; 42 2 50 59 58; 43 50 51 60 59; 44 51 52 61 60; 47 54 56 65 63;
48 56 57 66 65; 49 57 58 67 66; 50 58 59 68 67; 51 59 60 69 68; 52 60 61 70 69;
55 63 65 74 72; 56 65 66 75 74; 57 66 67 76 75; 58 67 68 77 76; 59 68 69 78 77;
60 69 70 79 78; 63 72 74 83 81; 64 74 75 84 83; 65 75 76 85 84; 66 76 77 86 85;
67 77 78 87 86; 68 78 79 88 87; 71 81 83 92 90; 72 83 84 93 92; 73 84 85 94 93;
74 85 86 95 94; 75 86 87 96 95; 76 87 88 97 96; 79 90 92 101 99;
80 92 93 102 101; 81 93 94 103 102; 82 94 95 104 103; 83 95 96 105 104;
84 96 97 106 105; 87 99 101 110 108; 88 101 102 111 110; 89 102 103 112 111;
90 103 104 113 112; 91 104 105 114 113; 92 105 106 115 114; 95 108 110 119 117;
96 110 111 120 119; 97 111 112 1 120; 98 112 113 121 1; 99 113 114 122 121;
100 114 115 123 122; 103 117 119 127 125; 104 119 120 128 127;
105 120 1 129 128; 106 1 121 130 129; 107 121 122 131 130; 108 122 123 132 131;
111 125 127 136 134; 112 127 128 137 136; 113 128 129 138 137;
114 129 130 139 138; 115 130 131 140 139; 116 131 132 141 140;
119 134 136 145 143; 120 136 137 146 145; 121 137 138 147 146;
122 138 139 148 147; 123 139 140 149 148; 124 140 141 150 149;
137 143 145 162 158; 140 145 146 164 162; 143 146 147 166 164;
146 147 148 168 166; 149 148 149 170 168; 152 149 150 172 170;
163 178 181 17 14; 166 181 183 19 17; 169 183 185 20 19; 172 185 187 22 20;
175 187 189 24 22; 178 189 191 26 24; 214 26 212 213 35; 217 35 213 216 44;
220 44 216 218 52; 223 52 218 220 61; 226 61 220 222 70; 229 70 222 224 79;
232 79 224 226 88; 235 88 226 228 97; 238 97 228 230 106; 241 106 230 232 115;
244 115 232 234 123; 247 123 234 236 132; 250 132 236 238 141;
253 141 238 240 150; 256 150 240 242 172; 265 191 247 212 26;
282 260 14 28 261; 285 261 28 37 263; 288 263 37 46 265; 291 265 46 54 267;
294 267 54 63 269; 297 269 63 72 271; 300 271 72 81 273; 303 273 81 90 275;
306 275 90 99 277; 309 277 99 108 279; 312 279 108 117 281;
315 281 117 125 283; 318 283 125 134 285; 321 285 134 143 287;
324 287 143 158 289; 333 295 178 14 260;
START GROUP DEFINITION
ELEMENT
_MX 15 16 19 20 23 24 27 28 31 32 35 36 39 40 43 44 47 48 51 52 55 56 59 60 -
63 64 67 68 71 72 75 76 79 80 83 84 87 88 91 92 95 96 99 100 103 104 107 -
108 111 112 115 116 119 120 123 124 137 140 149 152 163 166 175 178 214 217 -
220 223 226 229 232 235 238 241 244 247 250 253 256 265 282 285 288 291 294 -
297 300 303 306 309 312 315 318 321 324 333
_MY 15 TO 20 23 TO 28 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 TO 84 87 TO 92 -
111 TO 116 119 TO 124 137 140 143 146 149 152 163 166 169 172 175 178 214 -
217 226 229 232 235 238 241 250 253 256 265 282 285 294 297 300 303 306 309 -
318 321 324 333
_SHEAR 63 TO 68 71 TO 76 137 140 143 146 149 152 163 166 169 172 175 178 214 -
217 220 223 226 229 232 235 238 241 244 247 250 253 256 265 282 285 288 291 -
294 297 300 303 306 309 312 315 318 321 324 333
END GROUP DEFINITION
ELEMENT PROPERTY
15 TO 20 23 TO 28 31 TO 36 39 TO 44 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 -
80 TO 84 87 TO 92 95 TO 100 103 TO 108 111 TO 116 119 TO 124 137 140 143 146 -
```

```
149 152 163 166 169 172 175 178 214 217 220 223 226 229 232 235 238 241 244 -
247 250 253 256 265 282 285 288 291 294 297 300 303 306 309 312 315 318 321 -
324 333 THICKNESS 0.4
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 23.5616
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
1 TO 4 PRIS YD 0.6 ZD 0.6
5 PRIS YD 0.5 ZD 0.3
CONSTANTS
MATERIAL CONCRETE ALL
*****************************************
SUPPORTS
1 2 14 17 19 20 22 24 26 28 30 TO 35 37 39 TO 44 46 48 TO 52 54 56 TO 61 63 -
65 TO 70 72 74 TO 79 81 83 TO 88 90 92 TO 97 99 101 TO 106 108 110 TO 115 -
117 119 TO 123 125 127 TO 132 134 136 TO 141 143 145 TO 150 158 162 164 166 -
168 170 172 178 181 183 185 187 189 191 212 213 216 218 220 222 224 226 228 -
230 232 234 236 238 240 242 247 260 261 263 265 267 269 271 273 275 277 279 -
281 283 285 287 289 295 ELASTIC MAT DIRECT Y SUBGRADE 72000 PRINT
*****************************************
LOAD 1 DEAD LOAD    DL
SELFWEIGHT Y -1
JOINT LOAD
5 FX -0.001 FY -50.4 FZ 8
6 FX 0.001 FY -43.2 FZ 2.7
ELEMENT LOAD
15 TO 20 23 TO 28 31 TO 36 39 TO 44 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 -
80 TO 84 87 TO 92 95 TO 100 103 TO 108 111 TO 116 119 TO 124 137 140 143 146 -
149 152 163 166 169 172 175 178 214 217 220 223 226 229 232 235 238 241 244 -
247 250 253 256 265 282 285 288 291 294 297 300 303 306 309 312 315 318 321 -
324 333 PR GY -28.8
*****************************************
LOAD 2 LIVE LOAD    LL
JOINT LOAD
5 FX 0 FY -33.855 FZ -3.942
6 FX 0 FY -34.145 FZ 3.942
*****************************************
LOAD 3 MATERIAL LOAD    LL
JOINT LOAD
5 FX 0 FY -25.6 FZ -2.8
6 FX 0 FY -25.8 FZ 2.8
*****************************************
LOAD 4 WIND LOAD ALONG CONVEYOR WL (+Z)
JOINT LOAD
5 FX -0.003 FY 31.6 FZ 9.9
6 FX 0.003 FY -31.6 FZ 10.5
*****************************************
```

```
LOAD 5 WIND LOAD ALONG CONVEYOR WL (-Z)
JOINT LOAD
5 FX 0.003 FY -31.6 FZ -9.9
6 FX -0.003 FY 31.6 FZ -10.5
*****************************************
LOAD 6 WIND LOAD ACROSS THE CONV.  WL (+X)
JOINT LOAD
5 FX 0 FY 0 FZ 0
6 FX 0 FY 0 FZ 0
*****************************************
LOAD 7 WIND LOAD ACROSS THE CONV.   WL (-X)
JOINT LOAD
5 FX -0 FY 0 FZ 0
6 FX -0 FY 0 FZ 0
*****************************************
LOAD 8 TEMP LOAD    TL1
JOINT LOAD
5 FX 0 FY 0.1 FZ 22.4
6 FX 0 FY -0.1 FZ -22.4
*****************************************
LOAD 9 TEMP LOAD    TL2
JOINT LOAD
5 FX 0 FY -0.1 FZ -22.4
6 FX 0 FY 0.1 FZ 22.4
*****************************************
LOAD 10 SEISMIC LOAD ALONG THE CONV.    SL (+X)
JOINT LOAD
5 FX 1.958 FY 0.81 FZ 0.113
6 FX 1.958 FY 0.778 FZ -0.081
*****************************************
LOAD 11 SEISMIC LOAD ACROSS THE CONV.   SL (+Z)
JOINT LOAD
5 FX 0 FY 4.104 FZ 1.162
6 FX 0 FY -4.104 FZ 1.208
*****************************************
*****Load combination for bearing capacity check
LOAD 301 1.0 DL
REPEAT LOAD
1 1.0
LOAD 302 1.0 (DL + TL1) + 1.0  (LL+ML)
REPEAT LOAD
1 1.0 8 1.0 2 1.0 3 1.0
LOAD 303 1.0 (DL + TL1) + 1.0  (LL+ML)
REPEAT LOAD
1 1.0 9 1.0 2 1.0 3 1.0
LOAD 304 1.0 DL+ 0.75 WL(+Z) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 4 0.75 2 0.75 3 0.75
LOAD 305 1.0 DL+ 0.75 WL(-Z) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 5 0.75 2 0.75 3 0.75
LOAD 306 1.0 DL+ 0.75 WL(+X) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 6 0.75 2 0.75 3 0.75
```

```
LOAD 307 1.0 DL+ 0.75 WL(-X) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 7 0.75 2 0.75 3 0.75
LOAD 308 1.0 DL + 0.525 EL(+X) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 10 0.525 2 0.75 3 0.75
LOAD 309 1.0 DL + 0.525 EL(-X) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 10 -0.525 2 0.75 3 0.75
LOAD 310 1.0 DL + 0.525 EL(+Z) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 11 0.525 2 0.75 3 0.75
LOAD 311 1.0 DL + 0.525 EL(-Z) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 11 -0.525 2 0.75 3 0.75
LOAD 312 0.6 DL + 1.0 WL(+Z)
REPEAT LOAD
1 0.6 4 1.0
LOAD 313 0.6 DL + 1.0 WL(-Z)
REPEAT LOAD
1 0.6 5 1.0
LOAD 314 0.6 DL + 1.0 WL(+X)
REPEAT LOAD
1 0.6 6 1.0
LOAD 315 0.6 DL + 1.0 WL(-X)
REPEAT LOAD
1 0.6 7 1.0
LOAD 316 0.6 DL + 0.7 EL(+X)
REPEAT LOAD
1 0.6 10 0.7
LOAD 317 0.6 DL + 0.7 EL(-X)
REPEAT LOAD
1 0.6 10 -0.7
LOAD 318 0.6 DL + 0.7 EL(+Z)
REPEAT LOAD
1 0.6 11 0.7
LOAD 319 0.6 DL + 0.7 EL(-Z)
REPEAT LOAD
1 0.6 11 -0.7
*****************************************
*****Load combination for foundation design
LOAD 401 1.4 DL
REPEAT LOAD
1 1.4
LOAD 402 1.4 (DL + TL1) + 1.7  (LL+ML)
REPEAT LOAD
1 1.4 8 1.4 2 1.7 3 1.7
LOAD 403 1.4 (DL + TL1) + 1.7  (LL+ML)
REPEAT LOAD
1 1.4 9 1.4 2 1.7 3 1.7
LOAD 404 1.2 DL+ 1.6 WL(+Z) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 4 1.6 2 0.5 3 0.5
LOAD 405 1.2 DL+ 1.6 WL(-Z) + 0.5  (LL+ML)
```

```
REPEAT LOAD
1 1.2 5 1.6 2 0.5 3 0.5
LOAD 406 1.2 DL+ 1.6 WL(+X) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 6 1.6 2 0.5 3 0.5
LOAD 407 1.2 DL+ 1.6 WL(-X) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 7 1.6 2 0.5 3 0.5
LOAD 408 1.2 DL + 1.0 EL(+X) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 10 1.0 2 0.5 3 0.5
LOAD 409 1.2 DL + 1.0 EL(-X) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 10 -1.0 2 0.5 3 0.5
LOAD 410 1.2 DL + 1.0 EL(+Z) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 11 1.0 2 0.5 3 0.5
LOAD 411 1.2 DL + 1.0 EL(-Z) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 11 -1.0 2 0.5 3 0.5
LOAD 412 0.9 DL + 1.6 WL(+Z)
REPEAT LOAD
1 0.9 4 1.6
LOAD 413 0.9 DL + 1.6 WL(-Z)
REPEAT LOAD
1 0.9 5 1.6
LOAD 414 0.9 DL + 1.6 WL(+X)
REPEAT LOAD
1 0.9 6 1.6
LOAD 415 0.9 DL + 1.6 WL(-X)
REPEAT LOAD
1 0.9 7 1.6
LOAD 416 0.9 DL + 1.0 EL(+X)
REPEAT LOAD
1 0.9 10 1.0
LOAD 417 0.9 DL + 1.0 EL(-X)
REPEAT LOAD
1 0.9 10 -1.0
LOAD 418 0.9 DL + 1.0 EL(+Z)
REPEAT LOAD
1 0.9 11 1.0
LOAD 419 0.9 DL + 1.0 EL(-Z)
REPEAT LOAD
1 0.9 11 -1.0
****************************************
PERFORM ANALYSIS
****************************************
LOAD LIST 401 TO 419
START CONCRETE DESIGN
CODE ACI 2008
FC 35000 MEMB 1 TO 5
FYMAIN 420000 MEMB 1 TO 5
TRACK 1 ALL
CLB 0.05 MEMB 1 TO 5
```

```
CLT 0.05 MEMB 1 TO 5
CLS 0.05 ALL
MAXMAIN 25 ALL
MINMAIN 16 ALL
MINSEC 8 ALL
REINF 0 MEMB 1 TO 4
DESIGN COLUMN 1 TO 4
DESIGN BEAM 5
END CONCRETE DESIGN
FINISH
```

```
**************************************************
*                                                *
*           STAAD.Pro V8i SELECTseries6           *
*           Version  20.07.11.70                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    APR 16, 2020                *
*           Time=    15:32: 3                    *
*                                                *
*      USER ID:                                  *
**************************************************
```

```
    1. STAAD SPACE
INPUT FILE: D:\PROJECT\Maaden Gold\Conveyor\CONV. 3120-FDR-002\Staad\3120-FDR-002_TR1 FDN.STD
    2. START JOB INFORMATION
    3. ENGINEER DATE 15-APR-20
    4. JOB NAME MMGP-SAUDI ARABIA
    5. JOB CLIENT MA'ADEN
    6. JOB NO O19045/900707
    7. JOB REV 0
    8. JOB PART TRESTLE TR1 FOUNDATION
    9. JOB REF FOUNDATIONS FOR CONVEYOR 3120-FDR-002
   10. ENGINEER NAME APR
   11. CHECKER NAME SBY
   12. APPROVED NAME PRI
   13. END JOB INFORMATION
   14. INPUT WIDTH 79
   15. UNIT METER KN
   16. JOINT COORDINATES
   17. 1 0 -3 0; 2 0 -3 2; 3 0 -1 0; 4 0 -1 2; 5 0 -0.65 0; 6 0 -0.65 2
   18. 14 -0.75 -3 2.75; 17 -0.5 -3 2.75; 19 -0.25 -3 2.75; 20 0 -3 2.75
   19. 22 0.25 -3 2.75; 24 0.5 -3 2.75; 26 0.75 -3 2.75; 28 -0.75 -3 2.5
   20. 30 -0.5 -3 2.5; 31 -0.25 -3 2.5; 32 0 -3 2.5; 33 0.25 -3 2.5; 34 0.5 -3 2.5
   21. 35 0.75 -3 2.5; 37 -0.75 -3 2.25; 39 -0.5 -3 2.25; 40 -0.25 -3 2.25
   22. 41 0 -3 2.25; 42 0.25 -3 2.25; 43 0.5 -3 2.25; 44 0.75 -3 2.25; 46 -0.75 -3 2
   23. 48 -0.5 -3 2; 49 -0.25 -3 2; 50 0.25 -3 2; 51 0.5 -3 2; 52 0.75 -3 2
   24. 54 -0.75 -3 1.75; 56 -0.5 -3 1.75; 57 -0.25 -3 1.75; 58 0 -3 1.75
   25. 59 0.25 -3 1.75; 60 0.5 -3 1.75; 61 0.75 -3 1.75; 63 -0.75 -3 1.5
   26. 65 -0.5 -3 1.5; 66 -0.25 -3 1.5; 67 0 -3 1.5; 68 0.25 -3 1.5; 69 0.5 -3 1.5
   27. 70 0.75 -3 1.5; 72 -0.75 -3 1.25; 74 -0.5 -3 1.25; 75 -0.25 -3 1.25
   28. 76 0 -3 1.25; 77 0.25 -3 1.25; 78 0.5 -3 1.25; 79 0.75 -3 1.25; 81 -0.75 -3 1
   29. 83 -0.5 -3 1; 84 -0.25 -3 1; 85 0 -3 1; 86 0.25 -3 1; 87 0.5 -3 1
   30. 88 0.75 -3 1; 90 -0.75 -3 0.75; 92 -0.5 -3 0.75; 93 -0.25 -3 0.75
   31. 94 0 -3 0.75; 95 0.25 -3 0.75; 96 0.5 -3 0.75; 97 0.75 -3 0.75
   32. 99 -0.75 -3 0.5; 101 -0.5 -3 0.5; 102 -0.25 -3 0.5; 103 0 -3 0.5
   33. 104 0.25 -3 0.5; 105 0.5 -3 0.5; 106 0.75 -3 0.5; 108 -0.75 -3 0.25
   34. 110 -0.5 -3 0.25; 111 -0.25 -3 0.25; 112 0 -3 0.25; 113 0.25 -3 0.25
   35. 114 0.5 -3 0.25; 115 0.75 -3 0.25; 117 -0.75 -3 0; 119 -0.5 -3 0
   36. 120 -0.25 -3 0; 121 0.25 -3 0; 122 0.5 -3 0; 123 0.75 -3 0; 125 -0.75 -3 -0.25
   37. 127 -0.5 -3 -0.25; 128 -0.25 -3 -0.25; 129 0 -3 -0.25; 130 0.25 -3 -0.25
   38. 131 0.5 -3 -0.25; 132 0.75 -3 -0.25; 134 -0.75 -3 -0.5; 136 -0.5 -3 -0.5
```

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39. 137 -0.25 -3 -0.5; 138 0 -3 -0.5; 139 0.25 -3 -0.5; 140 0.5 -3 -0.5
40. 141 0.75 -3 -0.5; 143 -0.75 -3 -0.75; 145 -0.5 -3 -0.75; 146 -0.25 -3 -0.75
41. 147 0 -3 -0.75; 148 0.25 -3 -0.75; 149 0.5 -3 -0.75; 150 0.75 -3 -0.75
42. 158 -0.75 -3 -1; 162 -0.5 -3 -1; 164 -0.25 -3 -1; 166 0 -3 -1; 168 0.25 -3 -1.
43. 170 0.5 -3 -1; 172 0.75 -3 -1; 178 -0.75 -3 3; 181 -0.5 -3 3; 183 -0.25 -3 3
44. 185 0 -3 3; 187 0.25 -3 3; 189 0.5 -3 3; 191 0.75 -3 3; 212 1.00033 -3 2.75
45. 213 1.00033 -3 2.5; 216 1.00033 -3 2.25; 218 1.00033 -3 2; 220 1.00033 -3 1.75
46. 222 1.00033 -3 1.5; 224 1.00033 -3 1.25; 226 1.00033 -3 1; 228 1.00033 -3 0.75
47. 230 1.00033 -3 0.5; 232 1.00033 -3 0.25; 234 1.00033 -3 0
48. 236 1.00033 -3 -0.25; 238 1.00033 -3 -0.5; 240 1.00033 -3 -0.75
49. 242 1.00033 -3 -1; 247 1 -3 3; 260 -1.00033 -3 2.75; 261 -1.00033 -3 2.5
50. 263 -1.00033 -3 2.25; 265 -1.00033 -3 2; 267 -1.00033 -3 1.75; 269 -1 -3 1.5
51. 271 -1.00033 -3 1.25; 273 -1.00033 -3 1; 275 -1.00033 -3 0.75
52. 277 -1.00033 -3 0.5; 279 -1.00033 -3 0.25; 281 -1.00033 -3 0
53. 283 -1.00033 -3 -0.25; 285 -1.00033 -3 -0.5; 287 -1.00033 -3 -0.75
54. 289 -1.00033 -3 -1; 295 -1.00033 -3 3
55. MEMBER INCIDENCES
56. 1 1 3; 2 3 5; 3 2 4; 4 4 6; 5 3 4
57. ELEMENT INCIDENCES SHELL
58. 15 14 17 30 28; 16 17 19 31 30; 17 19 20 32 31; 18 20 22 33 32; 19 22 24 34 33
59. 20 24 26 35 34; 23 28 30 39 37; 24 30 31 40 39; 25 31 32 41 40; 26 32 33 42 41
60. 27 33 34 43 42; 28 34 35 44 43; 31 37 39 48 46; 32 39 40 49 48; 33 40 41 2 49
61. 34 41 42 50 2; 35 42 43 51 50; 36 43 44 52 51; 39 46 48 56 54; 40 48 49 57 56
62. 41 49 2 58 57; 42 2 50 59 58; 43 50 51 60 59; 44 51 52 61 60; 47 54 56 65 63
63. 48 56 57 66 65; 49 57 58 67 66; 50 58 59 68 67; 51 59 60 69 68; 52 60 61 70 69
64. 55 63 65 74 72; 56 65 66 75 74; 57 66 67 76 75; 58 67 68 77 76; 59 68 69 78 77
65. 60 69 70 79 78; 63 72 74 83 81; 64 74 75 84 83; 65 75 76 85 84; 66 76 77 86 85
66. 67 77 78 87 86; 68 78 79 88 87; 71 81 83 92 90; 72 83 84 93 92; 73 84 85 94 93
67. 74 85 86 95 94; 75 86 87 96 95; 76 87 88 97 96; 79 90 92 101 99
68. 80 92 93 102 101; 81 93 94 103 102; 82 94 95 104 103; 83 95 96 105 104
69. 84 96 97 106 105; 87 99 101 110 108; 88 101 102 111 110; 89 102 103 112 111
70. 90 103 104 113 112; 91 104 105 114 113; 92 105 106 115 114; 95 108 110 119 117
71. 96 110 111 120 119; 97 111 112 1 120; 98 112 113 121 1; 99 113 114 122 121
72. 100 114 115 123 122; 103 117 119 127 125; 104 119 120 128 127
73. 105 120 1 129 128; 106 1 121 130 129; 107 121 122 131 130; 108 122 123 132 131
74. 111 125 127 136 134; 112 127 128 137 136; 113 128 129 138 137
75. 114 129 130 139 138; 115 130 131 140 139; 116 131 132 141 140
76. 119 134 136 145 143; 120 136 137 146 145; 121 137 138 147 146
77. 122 138 139 148 147; 123 139 140 149 148; 124 140 141 150 149
78. 137 143 145 162 158; 140 145 146 164 162; 143 146 147 166 164
79. 146 147 148 168 166; 149 148 149 170 168; 152 149 150 172 170
80. 163 178 181 17 14; 166 181 183 19 17; 169 183 185 20 19; 172 185 187 22 20
81. 175 187 189 24 22; 178 189 191 26 24; 214 26 212 213 35; 217 35 213 216 44
82. 220 44 216 218 52; 223 52 218 220 61; 226 61 220 222 70; 229 70 222 224 79
83. 232 79 224 226 88; 235 88 226 228 97; 238 97 228 230 106; 241 106 230 232 115
84. 244 115 232 234 123; 247 123 234 236 132; 250 132 236 238 141
85. 253 141 238 240 150; 256 150 240 242 172; 265 191 247 212 26
86. 282 260 14 28 261; 285 261 28 37 263; 288 263 37 46 265; 291 265 46 54 267
87. 294 267 54 63 269; 297 269 63 72 271; 300 271 72 81 273; 303 273 81 90 275
88. 306 275 90 99 277; 309 277 99 108 279; 312 279 108 117 281
89. 315 281 117 125 283; 318 283 125 134 285; 321 285 134 143 287
90. 324 287 143 158 289; 333 295 178 14 260
91. START GROUP DEFINITION
92. ELEMENT
93. _MX 15 16 19 20 23 24 27 28 31 32 35 36 39 40 43 44 47 48 51 52 55 56 59 60 -
94. 63 64 67 68 71 72 75 76 79 80 83 84 87 88 91 92 95 96 99 100 103 104 107 -
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 95. 108 111 112 115 116 119 120 123 124 137 140 149 152 163 166 175 178 214 217 -
 96. 220 223 226 229 232 235 238 241 244 247 250 253 256 265 282 285 288 291 294 -
 97. 297 300 303 306 309 312 315 318 321 324 333
 98. _MY 15 TO 20 23 TO 28 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 TO 84 87 TO 92 -
 99. 111 TO 116 119 TO 124 137 140 143 146 149 152 163 166 169 172 175 178 214 -
100. 217 226 229 232 235 238 241 250 253 256 265 282 285 294 297 300 303 306 309 -
101. 318 321 324 333
102. _SHEAR 63 TO 68 71 TO 76 137 140 143 146 149 152 163 166 169 172 175 178 214 -
103. 217 220 223 226 229 232 235 238 241 244 247 250 253 256 265 282 285 288 291 -
104. 294 297 300 303 306 309 312 315 318 321 324 333
105. END GROUP DEFINITION
106. ELEMENT PROPERTY
107. 15 TO 20 23 TO 28 31 TO 36 39 TO 44 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 -
108. 80 TO 84 87 TO 92 95 TO 100 103 TO 108 111 TO 116 119 TO 124 137 140 143 146 -
109. 149 152 163 166 169 172 175 178 214 217 220 223 226 229 232 235 238 241 244 -
110. 247 250 253 256 265 282 285 288 291 294 297 300 303 306 309 312 315 318 321 -
111. 324 333 THICKNESS 0.4
112. DEFINE MATERIAL START
113. ISOTROPIC CONCRETE
114. E 2.17185E+007
115. POISSON 0.17
116. DENSITY 23.5616
117. ALPHA 1E-005
118. DAMP 0.05
119. TYPE CONCRETE
120. STRENGTH FCU 27579
121. END DEFINE MATERIAL
122. MEMBER PROPERTY AMERICAN
123. 1 TO 4 PRIS YD 0.6 ZD 0.6
124. 5 PRIS YD 0.5 ZD 0.3
125. CONSTANTS
126. MATERIAL CONCRETE ALL
127. ****************************************
128. SUPPORTS
129. 1 2 14 17 19 20 22 24 26 28 30 TO 35 37 39 TO 44 46 48 TO 52 54 56 TO 61 63 -
130. 65 TO 70 72 74 TO 79 81 83 TO 88 90 92 TO 97 99 101 TO 106 108 110 TO 115 -
131. 117 119 TO 123 125 127 TO 132 134 136 TO 141 143 145 TO 150 158 162 164 166 -
132. 168 170 172 178 181 183 185 187 189 191 212 213 216 218 220 222 224 226 228 -
133. 230 232 234 236 238 240 242 247 260 261 263 265 267 269 271 273 275 277 279 -
134. 281 283 285 287 289 295 ELASTIC MAT DIRECT Y SUBGRADE 72000 PRINT
```

JOINT	INFLUENCE AREA
1	0.062500
2	0.062500
14	0.062541
17	0.062500
19	0.062500
20	0.062500
22	0.062500
24	0.062500
26	0.062541
28	0.062541
30	0.062500
31	0.062500

32	0.062500
33	0.062500
34	0.062500
35	0.062541
37	0.062541
39	0.062500
40	0.062500
41	0.062500
42	0.062500
43	0.062500
44	0.062541
46	0.062541
48	0.062500
49	0.062500
50	0.062500
51	0.062500
52	0.062541
54	0.062541
56	0.062500
57	0.062500
58	0.062500
59	0.062500
60	0.062500
61	0.062541
63	0.062500
65	0.062500
66	0.062500
67	0.062500
68	0.062500
69	0.062500
70	0.062541
72	0.062541
74	0.062500
75	0.062500
76	0.062500
77	0.062500
78	0.062500
79	0.062541
81	0.062541
83	0.062500
84	0.062500
85	0.062500
86	0.062500
87	0.062500
88	0.062541
90	0.062541
92	0.062500
93	0.062500
94	0.062500
95	0.062500
96	0.062500
97	0.062541
99	0.062541
101	0.062500
102	0.062500
103	0.062500

104	0.062500
105	0.062500
106	0.062541
108	0.062541
110	0.062500
111	0.062500
112	0.062500
113	0.062500
114	0.062500
115	0.062541
117	0.062541
119	0.062500
120	0.062500
121	0.062500
122	0.062500
123	0.062541
125	0.062541
127	0.062500
128	0.062500
129	0.062500
130	0.062500
131	0.062500
132	0.062541
134	0.062541
136	0.062500
137	0.062500
138	0.062500
139	0.062500
140	0.062500
141	0.062541
143	0.062541
145	0.062500
146	0.062500
147	0.062500
148	0.062500
149	0.062500
150	0.062541
158	0.031271
162	0.031250
164	0.031250
166	0.031250
168	0.031250
170	0.031250
172	0.031271
178	0.031271
181	0.031250
183	0.031250
185	0.031250
187	0.031250
189	0.031250
191	0.031250
212	0.031271
213	0.031291
216	0.031291
218	0.031291
220	0.031291

222	0.031291
224	0.031291
226	0.031291
228	0.031291
230	0.031291
232	0.031291
234	0.031291
236	0.031291
238	0.031291
240	0.031291
242	0.015646
247	0.015646
260	0.031291
261	0.031291
263	0.031291
265	0.031291
267	0.033801
269	0.000041
271	0.033075
273	0.031291
275	0.031291
277	0.031291
279	0.031291
281	0.031291
283	0.031291
285	0.031291
287	0.031291
289	0.015646
295	0.015646
TOTAL	7.975605

```
135. ****************************************
136. LOAD 1 DEAD LOAD DL
137. SELFWEIGHT Y -1
138. JOINT LOAD
139. 5 FX -0.001 FY -50.4 FZ 8
140. 6 FX 0.001 FY -43.2 FZ 2.7
141. ELEMENT LOAD
142. 15 TO 20 23 TO 28 31 TO 36 39 TO 44 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 -
143. 80 TO 84 87 TO 92 95 TO 100 103 TO 108 111 TO 116 119 TO 124 137 140 143 146 -
144. 149 152 163 166 169 172 175 178 214 217 220 223 226 229 232 235 238 241 244 -
145. 247 250 253 256 265 282 285 288 291 294 297 300 303 306 309 312 315 318 321 -
146. 324 333 PR GY -28.8
147. ****************************************
148. LOAD 2 LIVE LOAD LL
149. JOINT LOAD
150. 5 FX 0 FY -33.855 FZ -3.942
151. 6 FX 0 FY -34.145 FZ 3.942
152. ****************************************
153. LOAD 3 MATERIAL LOAD LL
154. JOINT LOAD
155. 5 FX 0 FY -25.6 FZ -2.8
156. 6 FX 0 FY -25.8 FZ 2.8
157. ****************************************
158. LOAD 4 WIND LOAD ALONG CONVEYOR WL (+Z)
159. JOINT LOAD
160. 5 FX -0.003 FY 31.6 FZ 9.9
```

```
161. 6 FX 0.003 FY -31.6 FZ 10.5
162. ****************************************
163. LOAD 5 WIND LOAD ALONG CONVEYOR WL (-Z)
164. JOINT LOAD
165. 5 FX 0.003 FY -31.6 FZ -9.9
166. 6 FX -0.003 FY 31.6 FZ -10.5
167. ****************************************
168. LOAD 6 WIND LOAD ACROSS THE CONV.  WL (+X)
169. JOINT LOAD
170. 5 FX 0 FY 0 FZ 0
171. 6 FX 0 FY 0 FZ 0
172. ****************************************
173. LOAD 7 WIND LOAD ACROSS THE CONV.  WL (-X)
174. JOINT LOAD
175. 5 FX -0 FY 0 FZ 0
176. 6 FX -0 FY 0 FZ 0
177. ****************************************
178. LOAD 8 TEMP LOAD   TL1
179. JOINT LOAD
180. 5 FX 0 FY 0.1 FZ 22.4
181. 6 FX 0 FY -0.1 FZ -22.4
182. ****************************************
183. LOAD 9 TEMP LOAD   TL2
184. JOINT LOAD
185. 5 FX 0 FY -0.1 FZ -22.4
186. 6 FX 0 FY 0.1 FZ 22.4
187. ****************************************
188. LOAD 10 SEISMIC LOAD ALONG THE CONV. SL (+X)
189. JOINT LOAD
190. 5 FX 1.958 FY 0.81 FZ 0.113
191. 6 FX 1.958 FY 0.778 FZ -0.081
192. ****************************************
193. LOAD 11 SEISMIC LOAD ACROSS THE CONV. SL (+Z)
194. JOINT LOAD
195. 5 FX 0 FY 4.104 FZ 1.162
196. 6 FX 0 FY -4.104 FZ 1.208
197. ****************************************
198. *****LOAD COMBINATION FOR BEARING CAPACITY CHECK
199. LOAD 301 1.0 DL
200. REPEAT LOAD
201. 1 1.0
202. LOAD 302 1.0 (DL + TL1) + 1.0  (LL+ML)
203. REPEAT LOAD
204. 1 1.0 8 1.0 2 1.0 3 1.0
205. LOAD 303 1.0 (DL + TL1) + 1.0  (LL+ML)
206. REPEAT LOAD
207. 1 1.0 9 1.0 2 1.0 3 1.0
208. LOAD 304 1.0 DL+ 0.75 WL(+Z) + 0.75  (LL+ML)
209. REPEAT LOAD
210. 1 1.0 4 0.75 2 0.75 3 0.75
211. LOAD 305 1.0 DL+ 0.75 WL(-Z) + 0.75  (LL+ML)
212. REPEAT LOAD
213. 1 1.0 5 0.75 2 0.75 3 0.75
214. LOAD 306 1.0 DL+ 0.75 WL(+X) + 0.75  (LL+ML)
215. REPEAT LOAD
216. 1 1.0 6 0.75 2 0.75 3 0.75
```

```
217. LOAD 307 1.0 DL+ 0.75 WL(-X) + 0.75  (LL+ML)
218. REPEAT LOAD
219. 1 1.0 7 0.75 2 0.75 3 0.75
220. LOAD 308 1.0 DL + 0.525 EL(+X) + 0.75  (LL+ML)
221. REPEAT LOAD
222. 1 1.0 10 0.525 2 0.75 3 0.75
223. LOAD 309 1.0 DL + 0.525 EL(-X) + 0.75  (LL+ML)
224. REPEAT LOAD
225. 1 1.0 10 -0.525 2 0.75 3 0.75
226. LOAD 310 1.0 DL + 0.525 EL(+Z) + 0.75  (LL+ML)
227. REPEAT LOAD
228. 1 1.0 11 0.525 2 0.75 3 0.75
229. LOAD 311 1.0 DL + 0.525 EL(-Z) + 0.75  (LL+ML)
230. REPEAT LOAD
231. 1 1.0 11 -0.525 2 0.75 3 0.75
232. LOAD 312 0.6 DL + 1.0 WL(+Z)
233. REPEAT LOAD
234. 1 0.6 4 1.0
235. LOAD 313 0.6 DL + 1.0 WL(-Z)
236. REPEAT LOAD
237. 1 0.6 5 1.0
238. LOAD 314 0.6 DL + 1.0 WL(+X)
239. REPEAT LOAD
240. 1 0.6 6 1.0
241. LOAD 315 0.6 DL + 1.0 WL(-X)
242. REPEAT LOAD
243. 1 0.6 7 1.0
244. LOAD 316 0.6 DL + 0.7 EL(+X)
245. REPEAT LOAD
246. 1 0.6 10 0.7
247. LOAD 317 0.6 DL + 0.7 EL(-X)
248. REPEAT LOAD
249. 1 0.6 10 -0.7
250. LOAD 318 0.6 DL + 0.7 EL(+Z)
251. REPEAT LOAD
252. 1 0.6 11 0.7
253. LOAD 319 0.6 DL + 0.7 EL(-Z)
254. REPEAT LOAD
255. 1 0.6 11 -0.7
256. ****************************************
257. *****LOAD COMBINATION FOR FOUNDATION DESIGN
258. LOAD 401 1.4 DL
259. REPEAT LOAD
260. 1 1.4
261. LOAD 402 1.4 (DL + TL1) + 1.7  (LL+ML)
262. REPEAT LOAD
263. 1 1.4 8 1.4 2 1.7 3 1.7
264. LOAD 403 1.4 (DL + TL1) + 1.7  (LL+ML)
265. REPEAT LOAD
266. 1 1.4 9 1.4 2 1.7 3 1.7
267. LOAD 404 1.2 DL+ 1.6 WL(+Z) + 0.5  (LL+ML)
268. REPEAT LOAD
269. 1 1.2 4 1.6 2 0.5 3 0.5
270. LOAD 405 1.2 DL+ 1.6 WL(-Z) + 0.5  (LL+ML)
271. REPEAT LOAD
272. 1 1.2 5 1.6 2 0.5 3 0.5
```

```
273. LOAD 406 1.2 DL+ 1.6 WL(+X) + 0.5  (LL+ML)
274. REPEAT LOAD
275. 1 1.2 6 1.6 2 0.5 3 0.5
276. LOAD 407 1.2 DL+ 1.6 WL(-X) + 0.5  (LL+ML)
277. REPEAT LOAD
278. 1 1.2 7 1.6 2 0.5 3 0.5
279. LOAD 408 1.2 DL + 1.0 EL(+X) + 0.5  (LL+ML)
280. REPEAT LOAD
281. 1 1.2 10 1.0 2 0.5 3 0.5
282. LOAD 409 1.2 DL + 1.0 EL(-X) + 0.5  (LL+ML)
283. REPEAT LOAD
284. 1 1.2 10 -1.0 2 0.5 3 0.5
285. LOAD 410 1.2 DL + 1.0 EL(+Z) + 0.5  (LL+ML)
286. REPEAT LOAD
287. 1 1.2 11 1.0 2 0.5 3 0.5
288. LOAD 411 1.2 DL + 1.0 EL(-Z) + 0.5  (LL+ML)
289. REPEAT LOAD
290. 1 1.2 11 -1.0 2 0.5 3 0.5
291. LOAD 412 0.9 DL + 1.6 WL(+Z)
292. REPEAT LOAD
293. 1 0.9 4 1.6
294. LOAD 413 0.9 DL + 1.6 WL(-Z)
295. REPEAT LOAD
296. 1 0.9 5 1.6
297. LOAD 414 0.9 DL + 1.6 WL(+X)
298. REPEAT LOAD
299. 1 0.9 6 1.6
300. LOAD 415 0.9 DL + 1.6 WL(-X)
301. REPEAT LOAD
302. 1 0.9 7 1.6
303. LOAD 416 0.9 DL + 1.0 EL(+X)
304. REPEAT LOAD
305. 1 0.9 10 1.0
306. LOAD 417 0.9 DL + 1.0 EL(-X)
307. REPEAT LOAD
308. 1 0.9 10 -1.0
309. LOAD 418 0.9 DL + 1.0 EL(+Z)
310. REPEAT LOAD
311. 1 0.9 11 1.0
312. LOAD 419 0.9 DL + 1.0 EL(-Z)
313. REPEAT LOAD
314. 1 0.9 11 -1.0
315. ****************************************
316. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**157**	**NUMBER OF MEMBERS**	**5**
NUMBER OF PLATES	**128**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**153**

SOLVER USED IS THE IN-CORE ADVANCED MATH SOLVER

TOTAL PRIMARY LOAD CASES = 49, TOTAL DEGREES OF FREEDOM = 483
TOTAL LOAD COMBINATION CASES = 0 SO FAR.

317. **
318. LOAD LIST 401 TO 419
319. START CONCRETE DESIGN

```
320. CODE ACI 2008
321. FC 35000 MEMB 1 TO 5
322. FYMAIN 420000 MEMB 1 TO 5
323. TRACK 1 ALL
324. CLB 0.05 MEMB 1 TO 5
325. CLT 0.05 MEMB 1 TO 5
326. CLS 0.05 ALL
327. MAXMAIN 25 ALL
328. MINMAIN 16 ALL
329. MINSEC 8 ALL
330. REINF 0 MEMB 1 TO 4
331. DESIGN COLUMN 1 TO 4
```

==========

COLUMN NO. 1 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	109.35	0.01	0.00006

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	109.35	5.21	0.02380

==========

COLUMN NO. 2 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

```
 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  411.95  -1583.37    108.55      0.00    0.00000
 ---------------------------------------------------------

    COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  411.95  -1583.37    108.55      0.00    0.00000
 ---------------------------------------------------------
```

==========

COLUMN NO. 3 DESIGN PER ACI 318-08 - AXIAL + BENDING

```
FY - 420.0  FC -  35.0 MPA,  SQRE SIZE - 600.0 X 600.0 MMS, TIED
        ONLY MINIMUM STEEL IS REQUIRED.
        AREA OF STEEL REQUIRED =  3600.0  SQ. MM

BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
 ---------------------------------------------------------

 12 - 20 MM               1.047      401      END     0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER    8 SPACING 320.00 MM

    COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  411.95  -1583.37    120.26      0.01    0.00005
 ---------------------------------------------------------

    COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  411.95  -1583.37    120.26     13.14    0.05463
 ---------------------------------------------------------
```

==========

COLUMN NO. 4 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	93.05	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	93.05	0.00	0.00000

**********END OF COLUMN DESIGN RESULTS**********

332. DESIGN BEAM 5

BEAM NO. 5 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 2000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	2 - 16MM	0.	2000.	YES	YES

CRITICAL POS MOMENT=	21.98 KN-MET AT 0.MM, LOAD 404
REQD STEEL=	179.MM2, RHO=0.0014, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING=	230./ 41./ 168. MMS
REQD. DEVELOPMENT LENGTH =	410. MMS

Cracked Moment of Inertia Iz at above location = 41100.8 cm^4

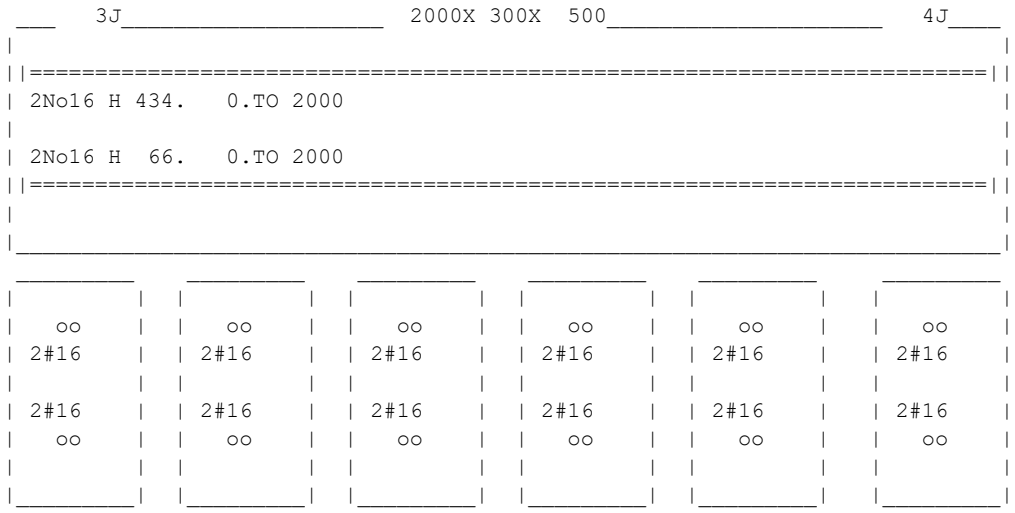
2	434.	2 - 16MM	0.	2000.	YES	YES

CRITICAL NEG MOMENT=	24.79 KN-MET AT 2000.MM, LOAD 404
REQD STEEL=	203.MM2, RHO=0.0016, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING=	230./ 41./ 168. MMS
REQD. DEVELOPMENT LENGTH =	410. MMS

Cracked Moment of Inertia Iz at above location = 41100.8 cm^4

B E A M N O. 5 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 5.78 KNS Vc= 123.46 KNS Vs= 0.00 KNS
Tu= 0.01 KN-MET Tc= 5.2 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 11.38 KNS Vc= 126.25 KNS Vs= 0.00 KNS
Tu= 0.01 KN-MET Tc= 5.2 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



**********END OF BEAM DESIGN**********

333. END CONCRETE DESIGN
334. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= APR 16,2020 TIME= 15:32: 8 ****

```
************************************************************
*   For technical assistance on STAAD.Pro, please visit    *
*   http://selectservices.bentley.com/en-US/               *
*                                                          *
*   Details about additional assistance from               *
*   Bentley and Partners can be found at program menu      *
*   Help->Technical Support                                *
*                                                          *
*      Copyright (c) 1997-2016 Bentley Systems, Inc.       *
*                http://www.bentley.com                    *
************************************************************
```

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2		
Date	16-04-2020	13-05-2020	31.10.2023				Date	16-04-2020	13-05-2020	31.10.2023		
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT		
Prepared By	APR	APR	APR				Prepared By	APR	APR	APR		
Checked	SBY	SBY	SBY				Checked	SBY	SBY	SBY		
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI		

ATTACHMENT 4 **STAAD INPUT & OUTPUT FILE OF FOUNDATION FOR TR-2**

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 15-Apr-20
JOB NAME MMGP-SAUDI ARABIA
JOB CLIENT MA'ADEN
JOB NO O19045/900707
JOB REV 0
JOB PART TRESTLE TR2 FOUNDATION
JOB REF FOUNDATIONS FOR CONVEYOR 3120-FDR-002
ENGINEER NAME APR
CHECKER NAME SBY
APPROVED NAME PRI
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
1 0 -3 0; 2 0 -3 2; 3 0 -1 0; 4 0 -1 2; 5 0 -0.65 0; 6 0 -0.65 2;
14 -0.75 -3 2.75; 17 -0.5 -3 2.75; 19 -0.25 -3 2.75; 20 0 -3 2.75;
22 0.25 -3 2.75; 24 0.5 -3 2.75; 26 0.75 -3 2.75; 28 -0.75 -3 2.5;
30 -0.5 -3 2.5; 31 -0.25 -3 2.5; 32 0 -3 2.5; 33 0.25 -3 2.5; 34 0.5 -3 2.5;
35 0.75 -3 2.5; 37 -0.75 -3 2.25; 39 -0.5 -3 2.25; 40 -0.25 -3 2.25;
41 0 -3 2.25; 42 0.25 -3 2.25; 43 0.5 -3 2.25; 44 0.75 -3 2.25; 46 -0.75 -3 2;
48 -0.5 -3 2; 49 -0.25 -3 2; 50 0.25 -3 2; 51 0.5 -3 2; 52 0.75 -3 2;
54 -0.75 -3 1.75; 56 -0.5 -3 1.75; 57 -0.25 -3 1.75; 58 0 -3 1.75;
59 0.25 -3 1.75; 60 0.5 -3 1.75; 61 0.75 -3 1.75; 63 -0.75 -3 1.5;
65 -0.5 -3 1.5; 66 -0.25 -3 1.5; 67 0 -3 1.5; 68 0.25 -3 1.5; 69 0.5 -3 1.5;
70 0.75 -3 1.5; 72 -0.75 -3 1.25; 74 -0.5 -3 1.25; 75 -0.25 -3 1.25;
76 0 -3 1.25; 77 0.25 -3 1.25; 78 0.5 -3 1.25; 79 0.75 -3 1.25; 81 -0.75 -3 1;
83 -0.5 -3 1; 84 -0.25 -3 1; 85 0 -3 1; 86 0.25 -3 1; 87 0.5 -3 1;
88 0.75 -3 1; 90 -0.75 -3 0.75; 92 -0.5 -3 0.75; 93 -0.25 -3 0.75;
94 0 -3 0.75; 95 0.25 -3 0.75; 96 0.5 -3 0.75; 97 0.75 -3 0.75;
99 -0.75 -3 0.5; 101 -0.5 -3 0.5; 102 -0.25 -3 0.5; 103 0 -3 0.5;
104 0.25 -3 0.5; 105 0.5 -3 0.5; 106 0.75 -3 0.5; 108 -0.75 -3 0.25;
110 -0.5 -3 0.25; 111 -0.25 -3 0.25; 112 0 -3 0.25; 113 0.25 -3 0.25;
114 0.5 -3 0.25; 115 0.75 -3 0.25; 117 -0.75 -3 0; 119 -0.5 -3 0;
120 -0.25 -3 0; 121 0.25 -3 0; 122 0.5 -3 0; 123 0.75 -3 0; 125 -0.75 -3 -0.25;
127 -0.5 -3 -0.25; 128 -0.25 -3 -0.25; 129 0 -3 -0.25; 130 0.25 -3 -0.25;
131 0.5 -3 -0.25; 132 0.75 -3 -0.25; 134 -0.75 -3 -0.5; 136 -0.5 -3 -0.5;
137 -0.25 -3 -0.5; 138 0 -3 -0.5; 139 0.25 -3 -0.5; 140 0.5 -3 -0.5;
141 0.75 -3 -0.5; 143 -0.75 -3 -0.75; 145 -0.5 -3 -0.75; 146 -0.25 -3 -0.75;
147 0 -3 -0.75; 148 0.25 -3 -0.75; 149 0.5 -3 -0.75; 150 0.75 -3 -0.75;
158 -0.75 -3 -1; 162 -0.5 -3 -1; 164 -0.25 -3 -1; 166 0 -3 -1; 168 0.25 -3 -1;
170 0.5 -3 -1; 172 0.75 -3 -1; 178 -0.75 -3 3; 181 -0.5 -3 3; 183 -0.25 -3 3;
185 0 -3 3; 187 0.25 -3 3; 189 0.5 -3 3; 191 0.75 -3 3; 212 1.00033 -3 2.75;
213 1.00033 -3 2.5; 216 1.00033 -3 2.25; 218 1.00033 -3 2; 220 1.00033 -3 1.75;
222 1.00033 -3 1.5; 224 1.00033 -3 1.25; 226 1.00033 -3 1; 228 1.00033 -3 0.75;
230 1.00033 -3 0.5; 232 1.00033 -3 0.25; 234 1.00033 -3 0;
236 1.00033 -3 -0.25; 238 1.00033 -3 -0.5; 240 1.00033 -3 -0.75;
242 1.00033 -3 -1; 247 1 -3 3; 260 -1.00033 -3 2.75; 261 -1.00033 -3 2.5;
263 -1.00033 -3 2.25; 265 -1.00033 -3 2; 267 -1 -3 1.75; 269 -1.00033 -3 1.5;
271 -1.00033 -3 1.25; 273 -1.00033 -3 1; 275 -1.00033 -3 0.75;
277 -1.00033 -3 0.5; 279 -1.00033 -3 0.25; 281 -1.00033 -3 0;
283 -1.00033 -3 -0.25; 285 -1.00033 -3 -0.5; 287 -1.00033 -3 -0.75;
289 -1.00033 -3 -1; 295 -1.00033 -3 3;
```

```
MEMBER INCIDENCES
1 1 3; 2 3 5; 3 2 4; 4 4 6; 5 3 4;
ELEMENT INCIDENCES SHELL
15 14 17 30 28; 16 17 19 31 30; 17 19 20 32 31; 18 20 22 33 32; 19 22 24 34 33;
20 24 26 35 34; 23 28 30 39 37; 24 30 31 40 39; 25 31 32 41 40; 26 32 33 42 41;
27 33 34 43 42; 28 34 35 44 43; 31 37 39 48 46; 32 39 40 49 48; 33 40 41 2 49;
34 41 42 50 2; 35 42 43 51 50; 36 43 44 52 51; 39 46 48 56 54; 40 48 49 57 56;
41 49 2 58 57; 42 2 50 59 58; 43 50 51 60 59; 44 51 52 61 60; 47 54 56 65 63;
48 56 57 66 65; 49 57 58 67 66; 50 58 59 68 67; 51 59 60 69 68; 52 60 61 70 69;
55 63 65 74 72; 56 65 66 75 74; 57 66 67 76 75; 58 67 68 77 76; 59 68 69 78 77;
60 69 70 79 78; 63 72 74 83 81; 64 74 75 84 83; 65 75 76 85 84; 66 76 77 86 85;
67 77 78 87 86; 68 78 79 88 87; 71 81 83 92 90; 72 83 84 93 92; 73 84 85 94 93;
74 85 86 95 94; 75 86 87 96 95; 76 87 88 97 96; 79 90 92 101 99;
80 92 93 102 101; 81 93 94 103 102; 82 94 95 104 103; 83 95 96 105 104;
84 96 97 106 105; 87 99 101 110 108; 88 101 102 111 110; 89 102 103 112 111;
90 103 104 113 112; 91 104 105 114 113; 92 105 106 115 114; 95 108 110 119 117;
96 110 111 120 119; 97 111 112 1 120; 98 112 113 121 1; 99 113 114 122 121;
100 114 115 123 122; 103 117 119 127 125; 104 119 120 128 127;
105 120 1 129 128; 106 1 121 130 129; 107 121 122 131 130; 108 122 123 132 131;
111 125 127 136 134; 112 127 128 137 136; 113 128 129 138 137;
114 129 130 139 138; 115 130 131 140 139; 116 131 132 141 140;
119 134 136 145 143; 120 136 137 146 145; 121 137 138 147 146;
122 138 139 148 147; 123 139 140 149 148; 124 140 141 150 149;
137 143 145 162 158; 140 145 146 164 162; 143 146 147 166 164;
146 147 148 168 166; 149 148 149 170 168; 152 149 150 172 170;
163 178 181 17 14; 166 181 183 19 17; 169 183 185 20 19; 172 185 187 22 20;
175 187 189 24 22; 178 189 191 26 24; 214 26 212 213 35; 217 35 213 216 44;
220 44 216 218 52; 223 52 218 220 61; 226 61 220 222 70; 229 70 222 224 79;
232 79 224 226 88; 235 88 226 228 97; 238 97 228 230 106; 241 106 230 232 115;
244 115 232 234 123; 247 123 234 236 132; 250 132 236 238 141;
253 141 238 240 150; 256 150 240 242 172; 265 191 247 212 26;
282 260 14 28 261; 285 261 28 37 263; 288 263 37 46 265; 291 265 46 54 267;
294 267 54 63 269; 297 269 63 72 271; 300 271 72 81 273; 303 273 81 90 275;
306 275 90 99 277; 309 277 99 108 279; 312 279 108 117 281;
315 281 117 125 283; 318 283 125 134 285; 321 285 134 143 287;
324 287 143 158 289; 333 295 178 14 260;
START GROUP DEFINITION
ELEMENT
_MX 15 16 19 20 23 24 27 28 31 32 35 36 39 40 43 44 47 48 51 52 55 56 59 60 -
63 64 67 68 71 72 75 76 79 80 83 84 87 88 91 92 95 96 99 100 103 104 107 -
108 111 112 115 116 119 120 123 124 137 140 149 152 163 166 175 178 214 217 -
220 223 226 229 232 235 238 241 244 247 250 253 256 265 282 285 288 291 294 -
297 300 303 306 309 312 315 318 321 324 333
_MY 15 TO 20 23 TO 28 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 TO 84 87 TO 92 -
111 TO 116 119 TO 124 137 140 143 146 149 152 163 166 169 172 175 178 214 -
217 226 229 232 235 238 241 250 253 256 265 282 285 294 297 300 303 306 309 -
318 321 324 333
_SHEAR 63 TO 68 71 TO 76 137 140 143 146 149 152 163 166 169 172 175 178 214 -
217 220 223 226 229 232 235 238 241 244 247 250 253 256 265 282 285 288 291 -
294 297 300 303 306 309 312 315 318 321 324 333
END GROUP DEFINITION
ELEMENT PROPERTY
15 TO 20 23 TO 28 31 TO 36 39 TO 44 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 -
80 TO 84 87 TO 92 95 TO 100 103 TO 108 111 TO 116 119 TO 124 137 140 143 146 -
```

```
149 152 163 166 169 172 175 178 214 217 220 223 226 229 232 235 238 241 244 -
247 250 253 256 265 282 285 288 291 294 297 300 303 306 309 312 315 318 321 -
324 333 THICKNESS 0.4
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 23.5616
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
1 TO 4 PRIS YD 0.6 ZD 0.6
5 PRIS YD 0.5 ZD 0.3
CONSTANTS
MATERIAL CONCRETE ALL
*****************************************
SUPPORTS
1 2 14 17 19 20 22 24 26 28 30 TO 35 37 39 TO 44 46 48 TO 52 54 56 TO 61 63 -
65 TO 70 72 74 TO 79 81 83 TO 88 90 92 TO 97 99 101 TO 106 108 110 TO 115 -
117 119 TO 123 125 127 TO 132 134 136 TO 141 143 145 TO 150 158 162 164 166 -
168 170 172 178 181 183 185 187 189 191 212 213 216 218 220 222 224 226 228 -
230 232 234 236 238 240 242 247 260 261 263 265 267 269 271 273 275 277 279 -
281 283 285 287 289 295 ELASTIC MAT DIRECT Y SUBGRADE 72000 PRINT
*****************************************
MEMBER OFFSET
5 START 0 -0.25 0
5 END 0 -0.25 0
LOAD 1 DEAD LOAD    DL
SELFWEIGHT Y -1
JOINT LOAD
5 FX -0.6 FY -51.4 FZ -0.7
6 FX -0.3 FY -49.8 FZ 7.1
ELEMENT LOAD
15 TO 20 23 TO 28 31 TO 36 39 TO 44 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 -
80 TO 84 87 TO 92 95 TO 100 103 TO 108 111 TO 116 119 TO 124 137 140 143 146 -
149 152 163 166 169 172 175 178 214 217 220 223 226 229 232 235 238 241 244 -
247 250 253 256 265 282 285 288 291 294 297 300 303 306 309 312 315 318 321 -
324 333 PR GY -28.8
*****************************************
LOAD 2 LIVE LOAD    LL
JOINT LOAD
5 FX -0.217 FY -32.1 FZ -2.458
6 FX -0.228 FY -31.9 FZ 2.458
*****************************************
LOAD 3 MATERIAL LOAD    LL
JOINT LOAD
5 FX -0.134 FY -25.6 FZ -1.549
6 FX -0.145 FY -25.4 FZ 1.549
*****************************************
LOAD 4 WIND LOAD ALONG CONVEYOR WL (+Z)
JOINT LOAD
```

```
5 FX 0.033 FY 43.6 FZ 12.7
6 FX -0.033 FY -43.6 FZ 13.0
*****************************************
LOAD 5 WIND LOAD ALONG CONVEYOR WL (-Z)
JOINT LOAD
5 FX -0.033 FY -43.7 FZ -12.7
6 FX 0.033 FY 43.7 FZ -13
*****************************************
LOAD 6 WIND LOAD ACROSS THE CONV.  WL (+X)
JOINT LOAD
5 FX 0.562 FY 0 FZ 0
6 FX 0.562 FY 0 FZ 0
*****************************************
LOAD 7 WIND LOAD ACROSS THE CONV.   WL (-X)
JOINT LOAD
5 FX -0.562 FY 0 FZ 0
6 FX -0.562 FY 0 FZ 0
*****************************************
LOAD 8 TEMP LOAD    TL1
JOINT LOAD
5 FX 0.028 FY 0.5 FZ -7.4
6 FX -0.028 FY -0.5 FZ 7.4
*****************************************
LOAD 9 TEMP LOAD    TL2
JOINT LOAD
5 FX -0.028 FY -0.5 FZ 7.4
6 FX 0.028 FY 0.5 FZ -7.4
*****************************************
LOAD 10 SEISMIC LOAD ALONG THE CONV.    SL (+X)
JOINT LOAD
5 FX 0.061 FY -0.5 FZ -0.035
6 FX 0.061 FY -0.5 FZ 0.035
*****************************************
LOAD 11 SEISMIC LOAD ACROSS THE CONV.   SL (+Z)
JOINT LOAD
5 FX 0 FY 8.5 FZ 2.7
6 FX 0 FY -8.5 FZ 2.8
*****************************************
*****Load combination for bearing capacity check
LOAD 301 1.0 DL
REPEAT LOAD
1 1.0
LOAD 302 1.0 (DL + TL1) + 1.0  (LL+ML)
REPEAT LOAD
1 1.0 8 1.0 2 1.0 3 1.0
LOAD 303 1.0 (DL + TL1) + 1.0  (LL+ML)
REPEAT LOAD
1 1.0 9 1.0 2 1.0 3 1.0
LOAD 304 1.0 DL+ 0.75 WL(+Z) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 4 0.75 2 0.75 3 0.75
LOAD 305 1.0 DL+ 0.75 WL(-Z) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 5 0.75 2 0.75 3 0.75
```

```
LOAD 306 1.0 DL+ 0.75 WL(+X) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 6 0.75 2 0.75 3 0.75
LOAD 307 1.0 DL+ 0.75 WL(-X) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 7 0.75 2 0.75 3 0.75
LOAD 308 1.0 DL + 0.525 EL(+X) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 10 0.525 2 0.75 3 0.75
LOAD 309 1.0 DL + 0.525 EL(-X) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 10 -0.525 2 0.75 3 0.75
LOAD 310 1.0 DL + 0.525 EL(+Z) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 11 0.525 2 0.75 3 0.75
LOAD 311 1.0 DL + 0.525 EL(-Z) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 11 -0.525 2 0.75 3 0.75
LOAD 312 0.6 DL + 1.0 WL(+Z)
REPEAT LOAD
1 0.6 4 1.0
LOAD 313 0.6 DL + 1.0 WL(-Z)
REPEAT LOAD
1 0.6 5 1.0
LOAD 314 0.6 DL + 1.0 WL(+X)
REPEAT LOAD
1 0.6 6 1.0
LOAD 315 0.6 DL + 1.0 WL(-X)
REPEAT LOAD
1 0.6 7 1.0
LOAD 316 0.6 DL + 0.7 EL(+X)
REPEAT LOAD
1 0.6 10 0.7
LOAD 317 0.6 DL + 0.7 EL(-X)
REPEAT LOAD
1 0.6 10 -0.7
LOAD 318 0.6 DL + 0.7 EL(+Z)
REPEAT LOAD
1 0.6 11 0.7
LOAD 319 0.6 DL + 0.7 EL(-Z)
REPEAT LOAD
1 0.6 11 -0.7
****************************************
*****Load combination for foundation design
LOAD 401 1.4 DL
REPEAT LOAD
1 1.4
LOAD 402 1.4 (DL + TL1) + 1.7  (LL+ML)
REPEAT LOAD
1 1.4 8 1.4 2 1.7 3 1.7
LOAD 403 1.4 (DL + TL1) + 1.7  (LL+ML)
REPEAT LOAD
1 1.4 9 1.4 2 1.7 3 1.7
LOAD 404 1.2 DL+ 1.6 WL(+Z) + 0.5  (LL+ML)
```

```
REPEAT LOAD
1 1.2 4 1.6 2 0.5 3 0.5
LOAD 405 1.2 DL+ 1.6 WL(-Z) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 5 1.6 2 0.5 3 0.5
LOAD 406 1.2 DL+ 1.6 WL(+X) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 6 1.6 2 0.5 3 0.5
LOAD 407 1.2 DL+ 1.6 WL(-X) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 7 1.6 2 0.5 3 0.5
LOAD 408 1.2 DL + 1.0 EL(+X) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 10 1.0 2 0.5 3 0.5
LOAD 409 1.2 DL + 1.0 EL(-X) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 10 -1.0 2 0.5 3 0.5
LOAD 410 1.2 DL + 1.0 EL(+Z) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 11 1.0 2 0.5 3 0.5
LOAD 411 1.2 DL + 1.0 EL(-Z) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 11 -1.0 2 0.5 3 0.5
LOAD 412 0.9 DL + 1.6 WL(+Z)
REPEAT LOAD
1 0.9 4 1.6
LOAD 413 0.9 DL + 1.6 WL(-Z)
REPEAT LOAD
1 0.9 5 1.6
LOAD 414 0.9 DL + 1.6 WL(+X)
REPEAT LOAD
1 0.9 6 1.6
LOAD 415 0.9 DL + 1.6 WL(-X)
REPEAT LOAD
1 0.9 7 1.6
LOAD 416 0.9 DL + 1.0 EL(+X)
REPEAT LOAD
1 0.9 10 1.0
LOAD 417 0.9 DL + 1.0 EL(-X)
REPEAT LOAD
1 0.9 10 -1.0
LOAD 418 0.9 DL + 1.0 EL(+Z)
REPEAT LOAD
1 0.9 11 1.0
LOAD 419 0.9 DL + 1.0 EL(-Z)
REPEAT LOAD
1 0.9 11 -1.0
*****************************************
PERFORM ANALYSIS
*****************************************
LOAD LIST 401 TO 419
START CONCRETE DESIGN
CODE ACI 2008
FC 35000 MEMB 1 TO 5
```

```
FYMAIN 420000 MEMB 1 TO 5
TRACK 1 ALL
CLB 0.05 MEMB 1 TO 5
CLT 0.05 MEMB 1 TO 5
CLS 0.05 ALL
MAXMAIN 20 MEMB 1 TO 4
MAXMAIN 12 MEMB 5
MINMAIN 16 MEMB 1 TO 4
MINMAIN 12 MEMB 5
MINSEC 8 ALL
REINF 0 MEMB 1 TO 4
DESIGN COLUMN 1 TO 4
DESIGN BEAM 5
END CONCRETE DESIGN
FINISH
```

```
**************************************************
*                                                *
*           STAAD.Pro V8i SELECTseries6           *
*           Version  20.07.11.70                  *
*           Proprietary Program of                *
*           Bentley Systems, Inc.                 *
*           Date=    APR 16, 2020                 *
*           Time=    15:41:30                     *
*                                                *
*      USER ID:                                  *
**************************************************
```

```
     1. STAAD SPACE
INPUT FILE: D:\PROJECT\Maaden Gold\Conveyor\CONV. 3120-FDR-002\Staad\3120-FDR-002_TR2 FDN.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 15-APR-20
     4. JOB NAME MMGP-SAUDI ARABIA
     5. JOB CLIENT MA'ADEN
     6. JOB NO O19045/900707
     7. JOB REV 0
     8. JOB PART TRESTLE TR2 FOUNDATION
     9. JOB REF FOUNDATIONS FOR CONVEYOR 3120-FDR-002
    10. ENGINEER NAME APR
    11. CHECKER NAME SBY
    12. APPROVED NAME PRI
    13. END JOB INFORMATION
    14. INPUT WIDTH 79
    15. UNIT METER KN
    16. JOINT COORDINATES
    17. 1 0 -3 0; 2 0 -3 2; 3 0 -1 0; 4 0 -1 2; 5 0 -0.65 0; 6 0 -0.65 2
    18. 14 -0.75 -3 2.75; 17 -0.5 -3 2.75; 19 -0.25 -3 2.75; 20 0 -3 2.75
    19. 22 0.25 -3 2.75; 24 0.5 -3 2.75; 26 0.75 -3 2.75; 28 -0.75 -3 2.5
    20. 30 -0.5 -3 2.5; 31 -0.25 -3 2.5; 32 0 -3 2.5; 33 0.25 -3 2.5; 34 0.5 -3 2.5
    21. 35 0.75 -3 2.5; 37 -0.75 -3 2.25; 39 -0.5 -3 2.25; 40 -0.25 -3 2.25
    22. 41 0 -3 2.25; 42 0.25 -3 2.25; 43 0.5 -3 2.25; 44 0.75 -3 2.25; 46 -0.75 -3 2
    23. 48 -0.5 -3 2; 49 -0.25 -3 2; 50 0.25 -3 2; 51 0.5 -3 2; 52 0.75 -3 2
    24. 54 -0.75 -3 1.75; 56 -0.5 -3 1.75; 57 -0.25 -3 1.75; 58 0 -3 1.75
    25. 59 0.25 -3 1.75; 60 0.5 -3 1.75; 61 0.75 -3 1.75; 63 -0.75 -3 1.5
    26. 65 -0.5 -3 1.5; 66 -0.25 -3 1.5; 67 0 -3 1.5; 68 0.25 -3 1.5; 69 0.5 -3 1.5
    27. 70 0.75 -3 1.5; 72 -0.75 -3 1.25; 74 -0.5 -3 1.25; 75 -0.25 -3 1.25
    28. 76 0 -3 1.25; 77 0.25 -3 1.25; 78 0.5 -3 1.25; 79 0.75 -3 1.25; 81 -0.75 -3 1
    29. 83 -0.5 -3 1; 84 -0.25 -3 1; 85 0 -3 1; 86 0.25 -3 1; 87 0.5 -3 1
    30. 88 0.75 -3 1; 90 -0.75 -3 0.75; 92 -0.5 -3 0.75; 93 -0.25 -3 0.75
    31. 94 0 -3 0.75; 95 0.25 -3 0.75; 96 0.5 -3 0.75; 97 0.75 -3 0.75
    32. 99 -0.75 -3 0.5; 101 -0.5 -3 0.5; 102 -0.25 -3 0.5; 103 0 -3 0.5
    33. 104 0.25 -3 0.5; 105 0.5 -3 0.5; 106 0.75 -3 0.5; 108 -0.75 -3 0.25
    34. 110 -0.5 -3 0.25; 111 -0.25 -3 0.25; 112 0 -3 0.25; 113 0.25 -3 0.25
    35. 114 0.5 -3 0.25; 115 0.75 -3 0.25; 117 -0.75 -3 0; 119 -0.5 -3 0
    36. 120 -0.25 -3 0; 121 0.25 -3 0; 122 0.5 -3 0; 123 0.75 -3 0; 125 -0.75 -3 -0.25
    37. 127 -0.5 -3 -0.25; 128 -0.25 -3 -0.25; 129 0 -3 -0.25; 130 0.25 -3 -0.25
    38. 131 0.5 -3 -0.25; 132 0.75 -3 -0.25; 134 -0.75 -3 -0.5; 136 -0.5 -3 -0.5
```

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39. 137 -0.25 -3 -0.5; 138 0 -3 -0.5; 139 0.25 -3 -0.5; 140 0.5 -3 -0.5
40. 141 0.75 -3 -0.5; 143 -0.75 -3 -0.75; 145 -0.5 -3 -0.75; 146 -0.25 -3 -0.75
41. 147 0 -3 -0.75; 148 0.25 -3 -0.75; 149 0.5 -3 -0.75; 150 0.75 -3 -0.75
42. 158 -0.75 -3 -1; 162 -0.5 -3 -1; 164 -0.25 -3 -1; 166 0 -3 -1; 168 0.25 -3 -1.
43. 170 0.5 -3 -1; 172 0.75 -3 -1; 178 -0.75 -3 3; 181 -0.5 -3 3; 183 -0.25 -3 3
44. 185 0 -3 3; 187 0.25 -3 3; 189 0.5 -3 3; 191 0.75 -3 3; 212 1.00033 -3 2.75
45. 213 1.00033 -3 2.5; 216 1.00033 -3 2.25; 218 1.00033 -3 2; 220 1.00033 -3 1.75
46. 222 1.00033 -3 1.5; 224 1.00033 -3 1.25; 226 1.00033 -3 1; 228 1.00033 -3 0.75
47. 230 1.00033 -3 0.5; 232 1.00033 -3 0.25; 234 1.00033 -3 0
48. 236 1.00033 -3 -0.25; 238 1.00033 -3 -0.5; 240 1.00033 -3 -0.75
49. 242 1.00033 -3 -1; 247 1 -3 3; 260 -1.00033 -3 2.75; 261 -1.00033 -3 2.5
50. 263 -1.00033 -3 2.25; 265 -1.00033 -3 2; 267 -1 -3 1.75; 269 -1.00033 -3 1.5
51. 271 -1.00033 -3 1.25; 273 -1.00033 -3 1; 275 -1.00033 -3 0.75
52. 277 -1.00033 -3 0.5; 279 -1.00033 -3 0.25; 281 -1.00033 -3 0
53. 283 -1.00033 -3 -0.25; 285 -1.00033 -3 -0.5; 287 -1.00033 -3 -0.75
54. 289 -1.00033 -3 -1; 295 -1.00033 -3 3
55. MEMBER INCIDENCES
56. 1 1 3; 2 3 5; 3 2 4; 4 4 6; 5 3 4
57. ELEMENT INCIDENCES SHELL
58. 15 14 17 30 28; 16 17 19 31 30; 17 19 20 32 31; 18 20 22 33 32; 19 22 24 34 33
59. 20 24 26 35 34; 23 28 30 39 37; 24 30 31 40 39; 25 31 32 41 40; 26 32 33 42 41
60. 27 33 34 43 42; 28 34 35 44 43; 31 37 39 48 46; 32 39 40 49 48; 33 40 41 2 49
61. 34 41 42 50 2; 35 42 43 51 50; 36 43 44 52 51; 39 46 48 56 54; 40 48 49 57 56
62. 41 49 2 58 57; 42 2 50 59 58; 43 50 51 60 59; 44 51 52 61 60; 47 54 56 65 63
63. 48 56 57 66 65; 49 57 58 67 66; 50 58 59 68 67; 51 59 60 69 68; 52 60 61 70 69
64. 55 63 65 74 72; 56 65 66 75 74; 57 66 67 76 75; 58 67 68 77 76; 59 68 69 78 77
65. 60 69 70 79 78; 63 72 74 83 81; 64 74 75 84 83; 65 75 76 85 84; 66 76 77 86 85
66. 67 77 78 87 86; 68 78 79 88 87; 71 81 83 92 90; 72 83 84 93 92; 73 84 85 94 93
67. 74 85 86 95 94; 75 86 87 96 95; 76 87 88 97 96; 79 90 92 101 99
68. 80 92 93 102 101; 81 93 94 103 102; 82 94 95 104 103; 83 95 96 105 104
69. 84 96 97 106 105; 87 99 101 110 108; 88 101 102 111 110; 89 102 103 112 111
70. 90 103 104 113 112; 91 104 105 114 113; 92 105 106 115 114; 95 108 110 119 117
71. 96 110 111 120 119; 97 111 112 1 120; 98 112 113 121 1; 99 113 114 122 121
72. 100 114 115 123 122; 103 117 119 127 125; 104 119 120 128 127
73. 105 120 1 129 128; 106 1 121 130 129; 107 121 122 131 130; 108 122 123 132 131
74. 111 125 127 136 134; 112 127 128 137 136; 113 128 129 138 137
75. 114 129 130 139 138; 115 130 131 140 139; 116 131 132 141 140
76. 119 134 136 145 143; 120 136 137 146 145; 121 137 138 147 146
77. 122 138 139 148 147; 123 139 140 149 148; 124 140 141 150 149
78. 137 143 145 162 158; 140 145 146 164 162; 143 146 147 166 164
79. 146 147 148 168 166; 149 148 149 170 168; 152 149 150 172 170
80. 163 178 181 17 14; 166 181 183 19 17; 169 183 185 20 19; 172 185 187 22 20
81. 175 187 189 24 22; 178 189 191 26 24; 214 26 212 213 35; 217 35 213 216 44
82. 220 44 216 218 52; 223 52 218 220 61; 226 61 220 222 70; 229 70 222 224 79
83. 232 79 224 226 88; 235 88 226 228 97; 238 97 228 230 106; 241 106 230 232 115
84. 244 115 232 234 123; 247 123 234 236 132; 250 132 236 238 141
85. 253 141 238 240 150; 256 150 240 242 172; 265 191 247 212 26
86. 282 260 14 28 261; 285 261 28 37 263; 288 263 37 46 265; 291 265 46 54 267
87. 294 267 54 63 269; 297 269 63 72 271; 300 271 72 81 273; 303 273 81 90 275
88. 306 275 90 99 277; 309 277 99 108 279; 312 279 108 117 281
89. 315 281 117 125 283; 318 283 125 134 285; 321 285 134 143 287
90. 324 287 143 158 289; 333 295 178 14 260
91. START GROUP DEFINITION
92. ELEMENT
93. _MX 15 16 19 20 23 24 27 28 31 32 35 36 39 40 43 44 47 48 51 52 55 56 59 60 -
94. 63 64 67 68 71 72 75 76 79 80 83 84 87 88 91 92 95 96 99 100 103 104 107 -
```

```
 95. 108 111 112 115 116 119 120 123 124 137 140 149 152 163 166 175 178 214 217 -
 96. 220 223 226 229 232 235 238 241 244 247 250 253 256 265 282 285 288 291 294 -
 97. 297 300 303 306 309 312 315 318 321 324 333
 98. _MY 15 TO 20 23 TO 28 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 TO 84 87 TO 92 -
 99. 111 TO 116 119 TO 124 137 140 143 146 149 152 163 166 169 172 175 178 214 -
100. 217 226 229 232 235 238 241 250 253 256 265 282 285 294 297 300 303 306 309 -
101. 318 321 324 333
102. _SHEAR 63 TO 68 71 TO 76 137 140 143 146 149 152 163 166 169 172 175 178 214 -
103. 217 220 223 226 229 232 235 238 241 244 247 250 253 256 265 282 285 288 291 -
104. 294 297 300 303 306 309 312 315 318 321 324 333
105. END GROUP DEFINITION
106. ELEMENT PROPERTY
107. 15 TO 20 23 TO 28 31 TO 36 39 TO 44 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 -
108. 80 TO 84 87 TO 92 95 TO 100 103 TO 108 111 TO 116 119 TO 124 137 140 143 146 -
109. 149 152 163 166 169 172 175 178 214 217 220 223 226 229 232 235 238 241 244 -
110. 247 250 253 256 265 282 285 288 291 294 297 300 303 306 309 312 315 318 321 -
111. 324 333 THICKNESS 0.4
112. DEFINE MATERIAL START
113. ISOTROPIC CONCRETE
114. E 2.17185E+007
115. POISSON 0.17
116. DENSITY 23.5616
117. ALPHA 1E-005
118. DAMP 0.05
119. TYPE CONCRETE
120. STRENGTH FCU 27579
121. END DEFINE MATERIAL
122. MEMBER PROPERTY AMERICAN
123. 1 TO 4 PRIS YD 0.6 ZD 0.6
124. 5 PRIS YD 0.5 ZD 0.3
125. CONSTANTS
126. MATERIAL CONCRETE ALL
127. ****************************************
128. SUPPORTS
129. 1 2 14 17 19 20 22 24 26 28 30 TO 35 37 39 TO 44 46 48 TO 52 54 56 TO 61 63 -
130. 65 TO 70 72 74 TO 79 81 83 TO 88 90 92 TO 97 99 101 TO 106 108 110 TO 115 -
131. 117 119 TO 123 125 127 TO 132 134 136 TO 141 143 145 TO 150 158 162 164 166 -
132. 168 170 172 178 181 183 185 187 189 191 212 213 216 218 220 222 224 226 228 -
133. 230 232 234 236 238 240 242 247 260 261 263 265 267 269 271 273 275 277 279 -
134. 281 283 285 287 289 295 ELASTIC MAT DIRECT Y SUBGRADE 72000 PRINT
```

JOINT	INFLUENCE AREA
1	0.062500
2	0.062500
14	0.062541
17	0.062500
19	0.062500
20	0.062500
22	0.062500
24	0.062500
26	0.062541
28	0.062541
30	0.062500
31	0.062500

32	0.062500
33	0.062500
34	0.062500
35	0.062541
37	0.062541
39	0.062500
40	0.062500
41	0.062500
42	0.062500
43	0.062500
44	0.062541
46	0.062541
48	0.062500
49	0.062500
50	0.062500
51	0.062500
52	0.062541
54	0.062500
56	0.062500
57	0.062500
58	0.062500
59	0.062500
60	0.062500
61	0.062541
63	0.062541
65	0.062500
66	0.062500
67	0.062500
68	0.062500
69	0.062500
70	0.062541
72	0.062541
74	0.062500
75	0.062500
76	0.062500
77	0.062500
78	0.062500
79	0.062541
81	0.062541
83	0.062500
84	0.062500
85	0.062500
86	0.062500
87	0.062500
88	0.062541
90	0.062541
92	0.062500
93	0.062500
94	0.062500
95	0.062500
96	0.062500
97	0.062541
99	0.062541
101	0.062500
102	0.062500
103	0.062500

104	0.062500
105	0.062500
106	0.062541
108	0.062541
110	0.062500
111	0.062500
112	0.062500
113	0.062500
114	0.062500
115	0.062541
117	0.062541
119	0.062500
120	0.062500
121	0.062500
122	0.062500
123	0.062541
125	0.062541
127	0.062500
128	0.062500
129	0.062500
130	0.062500
131	0.062500
132	0.062541
134	0.062541
136	0.062500
137	0.062500
138	0.062500
139	0.062500
140	0.062500
141	0.062541
143	0.062541
145	0.062500
146	0.062500
147	0.062500
148	0.062500
149	0.062500
150	0.062541
158	0.031271
162	0.031250
164	0.031250
166	0.031250
168	0.031250
170	0.031250
172	0.031271
178	0.031271
181	0.031250
183	0.031250
185	0.031250
187	0.031250
189	0.031250
191	0.031250
212	0.031271
213	0.031291
216	0.031291
218	0.031291
220	0.031291

222	0.031291
224	0.031291
226	0.031291
228	0.031291
230	0.031291
232	0.031291
234	0.031291
236	0.031291
238	0.031291
240	0.031291
242	0.015646
247	0.015646
260	0.031291
261	0.031291
263	0.031291
265	0.034042
267	0.000041
269	0.033316
271	0.031291
273	0.031291
275	0.031291
277	0.031291
279	0.031291
281	0.031291
283	0.031291
285	0.031291
287	0.031291
289	0.015646
295	0.015646
TOTAL	7.976088

```
135. ****************************************
136. MEMBER OFFSET
137. 5 START 0 -0.25 0
138. 5 END 0 -0.25 0
139. LOAD 1 DEAD LOAD DL
140. SELFWEIGHT Y -1
141. JOINT LOAD
142. 5 FX -0.6 FY -51.4 FZ -0.7
143. 6 FX -0.3 FY -49.8 FZ 7.1
144. ELEMENT LOAD
145. 15 TO 20 23 TO 28 31 TO 36 39 TO 44 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 -
146. 80 TO 84 87 TO 92 95 TO 100 103 TO 108 111 TO 116 119 TO 124 137 140 143 146 -
147. 149 152 163 166 169 172 175 178 214 217 220 223 226 229 232 235 238 241 244 -
148. 247 250 253 256 265 282 285 288 291 294 297 300 303 306 309 312 315 318 321 -
149. 324 333 PR GY -28.8
150. ****************************************
151. LOAD 2 LIVE LOAD LL
152. JOINT LOAD
153. 5 FX -0.217 FY -32.1 FZ -2.458
154. 6 FX -0.228 FY -31.9 FZ 2.458
155. ****************************************
156. LOAD 3 MATERIAL LOAD LL
157. JOINT LOAD
158. 5 FX -0.134 FY -25.6 FZ -1.549
159. 6 FX -0.145 FY -25.4 FZ 1.549
160. ****************************************
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161. LOAD 4 WIND LOAD ALONG CONVEYOR WL (+Z)
162. JOINT LOAD
163. 5 FX 0.033 FY 43.6 FZ 12.7
164. 6 FX -0.033 FY -43.6 FZ 13.0
165. ****************************************
166. LOAD 5 WIND LOAD ALONG CONVEYOR WL (-Z)
167. JOINT LOAD
168. 5 FX -0.033 FY -43.7 FZ -12.7
169. 6 FX 0.033 FY 43.7 FZ -13
170. ****************************************
171. LOAD 6 WIND LOAD ACROSS THE CONV.  WL (+X)
172. JOINT LOAD
173. 5 FX 0.562 FY 0 FZ 0
174. 6 FX 0.562 FY 0 FZ 0
175. ****************************************
176. LOAD 7 WIND LOAD ACROSS THE CONV.  WL (-X)
177. JOINT LOAD
178. 5 FX -0.562 FY 0 FZ 0
179. 6 FX -0.562 FY 0 FZ 0
180. ****************************************
181. LOAD 8 TEMP LOAD   TL1
182. JOINT LOAD
183. 5 FX 0.028 FY 0.5 FZ -7.4
184. 6 FX -0.028 FY -0.5 FZ 7.4
185. ****************************************
186. LOAD 9 TEMP LOAD   TL2
187. JOINT LOAD
188. 5 FX -0.028 FY -0.5 FZ 7.4
189. 6 FX 0.028 FY 0.5 FZ -7.4
190. ****************************************
191. LOAD 10 SEISMIC LOAD ALONG THE CONV. SL (+X)
192. JOINT LOAD
193. 5 FX 0.061 FY -0.5 FZ -0.035
194. 6 FX 0.061 FY -0.5 FZ 0.035
195. ****************************************
196. LOAD 11 SEISMIC LOAD ACROSS THE CONV. SL (+Z)
197. JOINT LOAD
198. 5 FX 0 FY 8.5 FZ 2.7
199. 6 FX 0 FY -8.5 FZ 2.8
200. ****************************************
201. *****LOAD COMBINATION FOR BEARING CAPACITY CHECK
202. LOAD 301 1.0 DL
203. REPEAT LOAD
204. 1 1.0
205. LOAD 302 1.0 (DL + TL1) + 1.0  (LL+ML)
206. REPEAT LOAD
207. 1 1.0 8 1.0 2 1.0 3 1.0
208. LOAD 303 1.0 (DL + TL1) + 1.0  (LL+ML)
209. REPEAT LOAD
210. 1 1.0 9 1.0 2 1.0 3 1.0
211. LOAD 304 1.0 DL+ 0.75 WL(+Z) + 0.75  (LL+ML)
212. REPEAT LOAD
213. 1 1.0 4 0.75 2 0.75 3 0.75
214. LOAD 305 1.0 DL+ 0.75 WL(-Z) + 0.75  (LL+ML)
215. REPEAT LOAD
216. 1 1.0 5 0.75 2 0.75 3 0.75
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217. LOAD 306 1.0 DL+ 0.75 WL(+X) + 0.75  (LL+ML)
218. REPEAT LOAD
219. 1 1.0 6 0.75 2 0.75 3 0.75
220. LOAD 307 1.0 DL+ 0.75 WL(-X) + 0.75  (LL+ML)
221. REPEAT LOAD
222. 1 1.0 7 0.75 2 0.75 3 0.75
223. LOAD 308 1.0 DL + 0.525 EL(+X) + 0.75  (LL+ML)
224. REPEAT LOAD
225. 1 1.0 10 0.525 2 0.75 3 0.75
226. LOAD 309 1.0 DL + 0.525 EL(-X) + 0.75  (LL+ML)
227. REPEAT LOAD
228. 1 1.0 10 -0.525 2 0.75 3 0.75
229. LOAD 310 1.0 DL + 0.525 EL(+Z) + 0.75  (LL+ML)
230. REPEAT LOAD
231. 1 1.0 11 0.525 2 0.75 3 0.75
232. LOAD 311 1.0 DL + 0.525 EL(-Z) + 0.75  (LL+ML)
233. REPEAT LOAD
234. 1 1.0 11 -0.525 2 0.75 3 0.75
235. LOAD 312 0.6 DL + 1.0 WL(+Z)
236. REPEAT LOAD
237. 1 0.6 4 1.0
238. LOAD 313 0.6 DL + 1.0 WL(-Z)
239. REPEAT LOAD
240. 1 0.6 5 1.0
241. LOAD 314 0.6 DL + 1.0 WL(+X)
242. REPEAT LOAD
243. 1 0.6 6 1.0
244. LOAD 315 0.6 DL + 1.0 WL(-X)
245. REPEAT LOAD
246. 1 0.6 7 1.0
247. LOAD 316 0.6 DL + 0.7 EL(+X)
248. REPEAT LOAD
249. 1 0.6 10 0.7
250. LOAD 317 0.6 DL + 0.7 EL(-X)
251. REPEAT LOAD
252. 1 0.6 10 -0.7
253. LOAD 318 0.6 DL + 0.7 EL(+Z)
254. REPEAT LOAD
255. 1 0.6 11 0.7
256. LOAD 319 0.6 DL + 0.7 EL(-Z)
257. REPEAT LOAD
258. 1 0.6 11 -0.7
259. ****************************************
260. *****LOAD COMBINATION FOR FOUNDATION DESIGN
261. LOAD 401 1.4 DL
262. REPEAT LOAD
263. 1 1.4
264. LOAD 402 1.4 (DL + TL1) + 1.7  (LL+ML)
265. REPEAT LOAD
266. 1 1.4 8 1.4 2 1.7 3 1.7
267. LOAD 403 1.4 (DL + TL1) + 1.7  (LL+ML)
268. REPEAT LOAD
269. 1 1.4 9 1.4 2 1.7 3 1.7
270. LOAD 404 1.2 DL+ 1.6 WL(+Z) + 0.5  (LL+ML)
271. REPEAT LOAD
272. 1 1.2 4 1.6 2 0.5 3 0.5
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```
273. LOAD 405 1.2 DL+ 1.6 WL(-Z) + 0.5  (LL+ML)
274. REPEAT LOAD
275. 1 1.2 5 1.6 2 0.5 3 0.5
276. LOAD 406 1.2 DL+ 1.6 WL(+X) + 0.5  (LL+ML)
277. REPEAT LOAD
278. 1 1.2 6 1.6 2 0.5 3 0.5
279. LOAD 407 1.2 DL+ 1.6 WL(-X) + 0.5  (LL+ML)
280. REPEAT LOAD
281. 1 1.2 7 1.6 2 0.5 3 0.5
282. LOAD 408 1.2 DL + 1.0 EL(+X) + 0.5  (LL+ML)
283. REPEAT LOAD
284. 1 1.2 10 1.0 2 0.5 3 0.5
285. LOAD 409 1.2 DL + 1.0 EL(-X) + 0.5  (LL+ML)
286. REPEAT LOAD
287. 1 1.2 10 -1.0 2 0.5 3 0.5
288. LOAD 410 1.2 DL + 1.0 EL(+Z) + 0.5  (LL+ML)
289. REPEAT LOAD
290. 1 1.2 11 1.0 2 0.5 3 0.5
291. LOAD 411 1.2 DL + 1.0 EL(-Z) + 0.5  (LL+ML)
292. REPEAT LOAD
293. 1 1.2 11 -1.0 2 0.5 3 0.5
294. LOAD 412 0.9 DL + 1.6 WL(+Z)
295. REPEAT LOAD
296. 1 0.9 4 1.6
297. LOAD 413 0.9 DL + 1.6 WL(-Z)
298. REPEAT LOAD
299. 1 0.9 5 1.6
300. LOAD 414 0.9 DL + 1.6 WL(+X)
301. REPEAT LOAD
302. 1 0.9 6 1.6
303. LOAD 415 0.9 DL + 1.6 WL(-X)
304. REPEAT LOAD
305. 1 0.9 7 1.6
306. LOAD 416 0.9 DL + 1.0 EL(+X)
307. REPEAT LOAD
308. 1 0.9 10 1.0
309. LOAD 417 0.9 DL + 1.0 EL(-X)
310. REPEAT LOAD
311. 1 0.9 10 -1.0
312. LOAD 418 0.9 DL + 1.0 EL(+Z)
313. REPEAT LOAD
314. 1 0.9 11 1.0
315. LOAD 419 0.9 DL + 1.0 EL(-Z)
316. REPEAT LOAD
317. 1 0.9 11 -1.0
318. ****************************************
319. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**157**	**NUMBER OF MEMBERS**	**5**
NUMBER OF PLATES	**128**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**153**

SOLVER USED IS THE IN-CORE ADVANCED MATH SOLVER

TOTAL PRIMARY LOAD CASES = 49, TOTAL DEGREES OF FREEDOM = 483
TOTAL LOAD COMBINATION CASES = 0 SO FAR.

320. **
321. LOAD LIST 401 TO 419
322. START CONCRETE DESIGN

```
323. CODE ACI 2008
324. FC 35000 MEMB 1 TO 5
325. FYMAIN 420000 MEMB 1 TO 5
326. TRACK 1 ALL
327. CLB 0.05 MEMB 1 TO 5
328. CLT 0.05 MEMB 1 TO 5
329. CLS 0.05 ALL
330. MAXMAIN 20 MEMB 1 TO 4
331. MAXMAIN 12 MEMB 5
332. MINMAIN 16 MEMB 1 TO 4
333. MINMAIN 12 MEMB 5
334. MINSEC 8 ALL
335. REINF 0 MEMB 1 TO 4
336. DESIGN COLUMN 1 TO 4
```

==========

COLUMN NO. 1 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	116.96	0.41	0.00176

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	116.96	6.74	0.02880

==========

COLUMN NO. 2 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	110.71	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	110.71	0.00	0.00000

==========

COLUMN NO. 3 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	129.02	0.27	0.00104

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	129.02	3.95	0.01529

==========

COLUMN NO. 4 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	107.26	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	107.26	0.00	0.00000

**********END OF COLUMN DESIGN RESULTS**********

337. DESIGN BEAM 5

BEAM NO. 5 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 2000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	64.	2 - 12MM	0.	2000.	YES	YES

CRITICAL POS MOMENT=	22.06 KN-MET AT 0.MM, LOAD 404
REQD STEEL=	179.MM2, RHO=0.0014, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING=	230./ 37./ 172. MMS
REQD. DEVELOPMENT LENGTH =	308. MMS

Cracked Moment of Inertia Iz at above location = 24968.6 cm^4

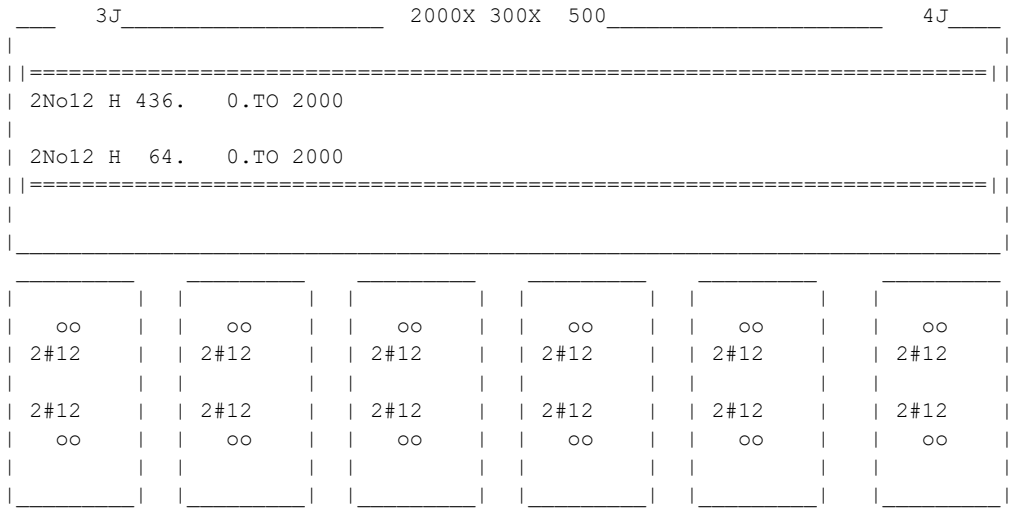
2	436.	2 - 12MM	0.	2000.	YES	YES

CRITICAL NEG MOMENT=	26.44 KN-MET AT 2000.MM, LOAD 404
REQD STEEL=	215.MM2, RHO=0.0016, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING=	230./ 37./ 172. MMS
REQD. DEVELOPMENT LENGTH =	308. MMS

Cracked Moment of Inertia Iz at above location = 24968.6 cm^4

B E A M N O. 5 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 2.24 KNS Vc= 127.36 KNS Vs= 0.00 KNS
Tu= 0.05 KN-MET Tc= 5.1 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 7.84 KNS Vc= 127.36 KNS Vs= 0.00 KNS
Tu= 0.05 KN-MET Tc= 5.1 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



**********END OF BEAM DESIGN**********

338. END CONCRETE DESIGN
339. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= APR 16,2020 TIME= 15:41:35 ****

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************************************************************
*   For technical assistance on STAAD.Pro, please visit    *
*   http://selectservices.bentley.com/en-US/               *
*                                                          *
*   Details about additional assistance from               *
*   Bentley and Partners can be found at program menu      *
*   Help->Technical Support                                *
*                                                          *
*      Copyright (c) 1997-2016 Bentley Systems, Inc.       *
*                http://www.bentley.com                    *
************************************************************
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